22

Heat Engines, Entropy, and the Second Law of Thermodynamics

CHAPTER OUTLINE

22.1 Heat Engines and the Second Law of Thermodynamics

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22.3 Reversible and Irreversible Processes

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22.5 Gasoline and Diesel Engines

22.6 Entropy

22.7 Changes in Entropy for Thermodynamic Systems

22.8 Entropy and the Second Law

* An asterisk indicates a question or problem new to this edition.

ANSWERS TO OBJECTIVE QUESTIONS

OQ22.1 Answer (d). The second law says that you must put in some work to pump heat from a lower-temperature to a higher-temperature location. But it can be very little work if the two temperatures are very nearly equal.

OQ22.2 Answer (d). Heat input will not *necessarily* produce an entropy increase, because a heat input could go on simultaneously with a larger work output, to carry the gas to a lower-temperature, lower-entropy final state. Work input will not *necessarily* produce an entropy increase, because work input could go on simultaneously with heat output to carry the gas to a lower-volume, lower-entropy final state. Either temperature increase at constant volume, or volume increase at constant temperature, or simultaneous increases in both temperature and volume, will necessarily end in a higher-entropy final state.

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OQ22.3 Answer (c). The coefficient of performance of this refrigerator is

$$\text{COP} = \frac{|Q_c|}{W} = \frac{115\ \text{kJ}}{18.0\ \text{kJ}} = 6.39$$

OQ22.4 Answer (c). Choice (c) is a statement of the first law of thermodynamics, *not* the second law. Choices (a), (b), (d), and (e) are alternative statements of the second law, (a) being the Kelvin-Planck formulation, (b) the Carnot statement, (d) the Clausius statement, and (e) summarizes the primary consequence of all these various statements.

OQ22.5 (i) Answer (b). (ii) Answer (a). (iii) Answer (b). (iv) Answer (a). (v) Answer (c). (vi) Answer (a). For any cyclic process the total input energy must be equal to the total output energy. This is a consequence of the first law of thermodynamics. It is satisfied by processes (ii), (iv), (v), and (vi) but not by processes (i) and (iii). The second law says that a cyclic process that takes in energy by heat must put out some of the energy by heat. This is not satisfied for process (v).

OQ22.6 Answer (a). The air conditioner operates on a cyclic process so the change in the internal energy of the refrigerant is zero. Then, the conservation of energy gives the thermal energy exhausted to the room as $Q_h = Q_c + W_{\text{eng}}$, where Q_c is the thermal energy the air conditioner removes from the room and W_{eng} is the work done to operate the device. Since $W_{\text{eng}} > 0$, the air conditioner is returning more thermal energy to the room than it is removing, so the average temperature in the room will increase.

OQ22.7 Answer (c). The maximum theoretical efficiency (the Carnot efficiency) of a device operating between absolute temperatures $T_c < T_h$ is $e_c = 1 - T_c/T_h$. For the given steam turbine, this is

$$e_c = 1 - \frac{3.0 \times 10^2\ \text{K}}{450\ \text{K}} = 0.33 \quad \text{or} \quad 33\%.$$

OQ22.8 Answer (d). The whole Universe must have an entropy change of zero or more. The environment around the system comprises the rest of the Universe, and must have an entropy change of +8.0 J/K, or more.

OQ22.9 Answer: $E > D > C > B > A$. Recall that for and ideal gas, $PV = nRT$, and $C_v = 3R/2$ and $C_p = 5R/2$.

Process A: isobaric, volume V goes to $0.5V$, so temperature T goes to $0.5T$, $dQ = nC_p dT$, so $dS = nC_p dT/T$; therefore $\Delta S = -\frac{5}{2} nR \ln 2$.

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Process B: isothermal, volume V goes to $0.5V$, so temperature T is constant and pressure P goes to $2P$, $dQ = pdV = nRT(dV/V)$, so $dS = nR(dV/V)$; therefore $\Delta S = -nR \ln 2$.

Process C: adiabatic, $Q = 0$; therefore $\Delta S = 0$.

Process D: isovolumetric, pressure P goes to $2P$, so temperature T goes to $2T$, $dQ = nC_v dT$, so $dS = n\,C_v\,dT/T$; therefore $\Delta S = \frac{3}{2} nR \ln 2$.

Process E: isobaric, volume V goes to $2V$; therefore $\Delta S = \frac{5}{2} nR \ln 2$.

OQ22.10 Answer (b). From conservation of energy, the energy input to the engine must be

$$Q_h = W_{\text{eng}} + Q_c = 15.0 \text{ kJ} + 37.0 \text{ kJ} = 52.0 \text{ kJ}$$

so the efficiency is

$$e = \frac{W_{\text{eng}}}{Q_c} = \frac{15.0 \text{ kJ}}{52.0 \text{ kJ}} = 0.288 \quad \text{or} \quad 28.8\%.$$

OQ22.11 Answer (b). In the reversible adiabatic expansion OA, the gas does work against a piston, takes in no energy by heat, and so drops in internal energy and in temperature. In the free adiabatic expansion OB, there is no piston, no work output, constant internal energy, and constant temperature for the ideal gas. Point A is at a lower temperature than O and point C is at an even lower temperature. The only point that could possibly have the same temperature as O is point B.

ANSWERS TO CONCEPTUAL QUESTIONS

CQ22.1 (a) The reduced flow rate of 'cooling water' reduces the amount of heat exhaust Q_c that the plant can put out each second. Even with constant efficiency, the rate at which the turbines can take in heat is reduced and so is the rate at which they can put out work to the generators. If anything, the efficiency will drop, because the smaller amount of water carrying the heat exhaust will tend to run hotter. The steam going through the turbines will undergo a smaller temperature change. Thus there are two reasons for the work output to drop.

(b) The engineer's version of events, as seen from inside the plant, is complete and correct. Hot steam pushes hard on the front of a turbine blade. Still-warm steam pushes less hard on the back of

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the blade, which turns in response to the pressure difference. Higher temperature at the heat exhaust port in the lake works its way back to a corresponding higher temperature of the steam leaving a turbine blade, a smaller temperature drop across the blade, and a lower work output.

CQ22.2 One: Energy flows by heat from a hot bowl of chili into the cooler surrounding air. Heat lost by the hot stuff is equal to heat gained by the cold stuff, but the entropy decrease of the hot stuff is less than the entropy increase of the cold stuff.

Two: As you inflate a soft car tire at a service station, air from a tank at high pressure expands to fill a larger volume. That air increases in entropy and the surrounding atmosphere undergoes no significant entropy change.

Three: The brakes of your car get warm as you come to a stop. The shoes and drums increase in entropy and nothing loses energy by heat, so nothing decreases in entropy.

CQ22.3 No. The first law of thermodynamics is a statement about energy conservation, while the second is a statement about stable thermal equilibrium. They are by no means mutually exclusive. For the particular case of a cycling heat engine, the first law implies $|Q_h| = W_{\text{eng}} + |Q_c|$, and the second law implies $|Q_c| > 0$.

CQ22.4 Take an automobile as an example. According to the first law or the idea of energy conservation, it must take in all the energy it puts out. Its energy source is chemical energy in gasoline. During the combustion process, some of that energy goes into moving the pistons and eventually into the mechanical motion of the car. The chemical potential energy turning into internal energy can be modeled as energy input by heat. The second law says that not all of the energy input can become output mechanical energy. Much of the input energy must and does become energy output by heat, which, through the cooling system, is dissipated into the atmosphere. Moreover, there are numerous places where friction, both mechanical and fluid, turns mechanical energy into internal energy. In even the most efficient internal combustion engine cars, less than 30% of the energy from the fuel actually goes into moving the car. The rest ends up as useless internal energy in the atmosphere.

CQ22.5 Either statement can be considered an instructive analogy. We choose to take the first view. All processes require energy, either as energy content or as energy input. The kinetic energy that it possessed at its formation continues to make the Earth go around. Energy released by nuclear reactions in the core of the Sun drives weather on the Earth and essentially all processes in the biosphere.

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The energy intensity of sunlight controls how lush a forest or jungle can be and how warm a planet is. Continuous energy input is not required for the motion of the planet. Continuous energy input is required for life because energy tends to be continuously degraded, as heat flows into lower-temperature sinks. The continuously increasing entropy of the Universe is the index to energy-transfers completed.

CQ22.6 (a) A slice of hot pizza cools off. Road friction brings a skidding car to a stop. A cup falls to the floor and shatters. Your cat dies. Any process is irreversible if it looks funny or frightening when shown in a videotape running backwards. (b) The free flight of a projectile is nearly reversible.

CQ22.7 (a) When the two sides of the semiconductor are at different temperatures, an electric potential (voltage) is generated across the material, which can drive electric current through an external circuit. The two cups at 50°C contain the same amount of internal energy as the pair of hot and cold cups. But no energy flows by heat through the converter bridging between them and no voltage is generated across the semiconductors.

(b) A heat engine must put out exhaust energy by heat. The cold cup provides a sink to absorb output or wasted energy by heat, which has nowhere to go between two cups of equally warm water.

CQ22.8 A higher steam temperature means that more energy can be extracted from the steam. For a constant temperature heat sink at T_c, and steam at T_h, the efficiency of the power plant goes as $\frac{T_h - T_c}{T_h} = 1 - \frac{T_c}{T_h}$ and is maximized for a high T_h.

CQ22.9 (a) For an expanding ideal gas at constant temperature, the internal energy stays constant. The gas must absorb by heat the same amount of energy that it puts out by work. Then its entropy change is $\Delta S = \frac{\Delta Q}{T} = nR\ln\left(\frac{V_2}{V_1}\right)$.

(b) For a reversible adiabatic expansion, $\Delta Q = 0$ and $\Delta S = 0$. An ideal gas undergoing an irreversible adiabatic expansion can have any positive value for ΔS up to the value given in part (a).

CQ22.10 No. Your roommate creates "order" locally, but as she works, she transfers energy by heat to the room, causing the net entropy to increase. An analogy used by Carnot is instructive: A waterfall continuously converts mechanical energy into internal energy. It

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continuously creates entropy as the motion of the falling water turns into molecular motion at the bottom of the falls. We humans put turbines into the waterfall, diverting some of the energy stream to our use. Water flows spontaneously from high to low elevation and energy spontaneously flows by heat from high to low temperature. Into the great flow of solar radiation from Sun to Earth, living things put themselves. They live on energy flow, more than just on energy. A basking snake diverts energy from a high-temperature source (the Sun) through itself temporarily, before the energy inevitably is radiated from the body of the snake to a low-temperature sink (outer space). A tree builds cellulose molecules and we build libraries and babies who look like their grandmothers, all out of a thin diverted stream in the universal flow of energy. We do not violate the second law, for we build local reductions in the entropy of one thing within the inexorable increase in the total entropy of the Universe.

CQ22.11 No. An engine with no thermal pollution would absorb energy from a reservoir and convert it completely into work; this is a clear violation of the second law of thermodynamics.

CQ22.12 (a) Shaking opens up spaces between jellybeans. The smaller ones more often can fall down into spaces below them. (b) The accumulation of larger candies on top and smaller ones on the bottom implies a small decrease in one contribution to the total entropy, but the second law is not violated. The total entropy increases as the system warms up, its increase in internal energy coming from the work put into shaking the box and also from a small decrease in gravitational potential energy as the beans settle compactly together.

CQ22.13 First, the efficiency of the automobile engine cannot exceed the Carnot efficiency: it is limited by the temperature of burning fuel and the temperature of the environment into which the exhaust is dumped. Second, the engine block cannot be allowed to go over a certain temperature. Third, any practical engine has friction, incomplete burning of fuel, and limits set by timing and energy transfer by heat.

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SOLUTIONS TO END-OF-CHAPTER PROBLEMS

Section 22.1 Heat Engines and the Second Law of Thermodynamics

P22.1 (a) We have $e = \dfrac{W_{\text{eng}}}{|Q_h|} = 1 - \dfrac{|Q_c|}{|Q_h|} \quad \rightarrow \quad \dfrac{|Q_c|}{|Q_h|} = 1 - e \quad \rightarrow \quad |Q_h| = \dfrac{|Q_c|}{1-e}$

With $|Q_c| = 8\ 000\ \text{J},$ we have $|Q_h| = \dfrac{|Q_c|}{1-e} = \dfrac{8\ 000\ \text{J}}{1 - 0.250} = \boxed{10.7\ \text{kJ}}$

(b) The work per cycle is

$$W_{\text{eng}} = |Q_h| - |Q_c| = 2\ 667\ \text{J}$$

From the definition of output power,

$$P = \frac{W_{\text{eng}}}{\Delta t}$$

we have the time for one cycle:

$$\Delta t = \frac{W_{\text{eng}}}{P} = \frac{2\ 667\ \text{J}}{5\ 000\ \text{J/s}} = \boxed{0.533\ \text{s}}$$

P22.2 (a) The efficiency of a heat engine is $e = W_{\text{env}}/|Q_h|$, where W_{env} is the work done by the engine and $|Q_h|$ is the energy absorbed from the higher temperature reservoir. Thus, if $W_{\text{env}} = |Q_h|/4$, the efficiency is $e = 1/4 = \boxed{0.25 \text{ or } 25\%}$.

(b) From conservation of energy, the energy exhausted to the lower temperature reservoir is $|Q_c| = |Q_h| - W_{\text{env}}$. Therefore, if $W_{\text{env}} = |Q_h|/4$, we have $|Q_c| = 3|Q_h|/4$ or $\boxed{|Q_c|/|Q_h| = 3/4}$.

P22.3 (a) The efficiency of the engine is

$$e = \frac{W_{\text{eng}}}{|Q_h|} = \frac{25.0\ \text{J}}{360\ \text{J}} = \boxed{0.069\ 4} \text{ or } \boxed{6.94\%}$$

(b) The energy expelled to the cold reservoir during each cycle is

$$|Q_c| = |Q_h| - W_{\text{eng}} = 360\ \text{J} - 25.0\ \text{J} = \boxed{335\ \text{J}}$$

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P22.4 The engine's output work we identify with the kinetic energy of the bullet:

$$W_{\text{eng}} = K = \frac{1}{2}mv^2 = \frac{1}{2}(0.002\ 4\ \text{kg})(320\ \text{m/s})^2 = 123\ \text{J}$$

$$e = \frac{W_{\text{eng}}}{Q_h}$$

$$Q_h = \frac{W_{\text{eng}}}{e} = \frac{123\ \text{J}}{0.011} = 1.12 \times 10^4\ \text{J}$$

$$Q_h = W_{\text{eng}} + |Q_c|$$

The energy exhaust is

$$|Q_c| = Q_h - W_{\text{eng}} = 1.12 \times 10^4\ \text{J} - 123\ \text{J} = 1.10 \times 10^4\ \text{J}$$

$$Q = mc\Delta T$$

$$\Delta T = \frac{Q}{mc} = \frac{1.10 \times 10^4\ \text{J} \cdot \text{kg°C}}{(1.80\ \text{kg})(448\ \text{J})} = \boxed{13.7°\text{C}}$$

P22.5 (a) The engine's efficiency is given by

$$e = \frac{W_{\text{eng}}}{|Q_h|} = \frac{|Q_h| - |Q_c|}{|Q_h|} = 1 - \frac{|Q_c|}{|Q_h|} = 1 - \frac{1.20\ \text{kJ}}{1.70\ \text{kJ}}$$

$$= \boxed{0.294\ (\text{or } 29.4\%)}$$

(b) During each cycle, the work done by the engine is

$$W_{\text{eng}} = |Q_h| - |Q_c| = 1.70\ \text{kJ} - 1.20\ \text{kJ} = \boxed{5.00 \times 10^2\ \text{J}}$$

(c) The power transferred out of the engine is

$$P = \frac{W_{\text{eng}}}{\Delta t} = \frac{5.00 \times 10^2\ \text{J}}{0.300\ \text{s}} = 1.67 \times 10^3\ \text{W} = \boxed{1.67\ \text{kW}}$$

P22.6 (a) The input energy each hour is

$$(7.89 \times 10^3\ \text{J/revolution})(2\ 500\ \text{rev/min})\left(\frac{60\ \text{min}}{1\ \text{h}}\right)$$

$$= 1.18 \times 10^9\ \text{J/h}$$

implying fuel input $(1.18 \times 10^9\ \text{J/h})\left(\frac{1\ \text{L}}{4.03 \times 10^7\ \text{J}}\right) = \boxed{29.4\ \text{L/h}}$.

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(b) $Q_h = W_{\text{eng}} + |Q_c|$. For a continuous-transfer process we may divide by time to have

$$\frac{Q_h}{\Delta t} = \frac{W_{\text{eng}}}{\Delta t} + \frac{|Q_c|}{\Delta t}$$

$$\text{Useful power output} = \frac{W_{\text{eng}}}{\Delta t} = \frac{Q_h}{\Delta t} - \frac{|Q_c|}{\Delta t}$$

$$= \left(\frac{7.89\times10^3\ \text{J}}{\text{revolution}} - \frac{4.58\times10^3\ \text{J}}{\text{revolution}}\right)\left(\frac{2\ 500\ \text{rev}}{1\ \text{min}}\right)\left(\frac{1\ \text{min}}{60\ \text{s}}\right)$$

$$= 1.38\times10^5\ \text{W}$$

$$P_{\text{eng}} = \left(1.38\times10^5\ \text{W}\right)\left(\frac{1\ \text{hp}}{746\ \text{W}}\right) = \boxed{185\ \text{hp}}$$

(c) $$P_{\text{eng}} = \tau\omega \Rightarrow \tau = \frac{P_{\text{eng}}}{\omega} = \left(\frac{1.38\times10^5\ \text{J/s}}{2\ 500\ \text{rev/60 s}}\right)\left(\frac{1\ \text{rev}}{2\pi\ \text{rad}}\right) = \boxed{527\ \text{N}\cdot\text{m}}$$

(d) $$\frac{|Q_c|}{\Delta t} = \left(\frac{4.58\times10^3\ \text{J}}{\text{revolution}}\right)\left(\frac{2\ 500\ \text{rev}}{60\ \text{s}}\right) = \boxed{1.91\times10^5\ \text{W}}$$

P22.7 The energy to melt a mass Δm_{Hg} of Hg is $|Q_c| = m_{\text{Hg}} L_f$. The energy absorbed to freeze Δm_{Al} of aluminum is $|Q_h| = m_{\text{Al}} L_f$. The efficiency is

$$e = 1 - \frac{|Q_c|}{|Q_h|} = 1 - \frac{\Delta m_{\text{Hg}} L_{\text{Hg}}}{\Delta m_{\text{Al}} L_{\text{Al}}} = 1 - \frac{(15.0\ \text{g})(1.18\times10^4\ \text{J/kg})}{(1.00\ \text{g})(3.97\times10^5\ \text{J/kg})}$$

$$= 0.554 = \boxed{55.4\%}$$

Section 22.2 Heat Pumps and Refrigerators

P22.8 $$\text{COP}(\text{refrigerator}) = \frac{|Q_c|}{W}$$

(a) If $|Q_c| = 120\ \text{J}$ and COP = 5.00, then $W = \boxed{24.0\ \text{J}}$.

(b) $|Q_h| = |Q_c| + W = 120\ \text{J} + 24\ \text{J} = \boxed{144\ \text{J}}$

P22.9 (a) The work done on the refrigerant in each cycle is

$$W = Q_H - Q_L = 625\ \text{kJ} - 550\ \text{kJ} = \boxed{75.0\ \text{kJ}}$$

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(b) The coefficient of performance of a refrigerator is:

$$\text{COP} = \frac{Q_L}{W} = \frac{Q_L}{Q_H - Q_L}$$

Solving numerically:

$$\text{COP} = \frac{Q_L}{W} = \frac{Q_L}{Q_H - Q_L} = \frac{550\ \text{kJ}}{625\ \text{kJ} - 550\ \text{kJ}} = \boxed{7.33}$$

P22.10 (a) The coefficient of performance of a heat pump is $\text{COP} = |Q_h|/W$, where $|Q_h|$ is the thermal energy delivered to the warm space and W is the work input required to operate the heat pump. Therefore,

$$|Q_h| = W \cdot \text{COP} = (P \cdot \Delta t) \cdot \text{COP}$$

$$= \left[\left(7.03 \times 10^3 \ \frac{\text{J}}{\cancel{\text{s}}} \right) \left(8.00 \ \cancel{\text{h}} \right) \left(\frac{3\ 600 \ \cancel{\text{s}}}{1 \ \cancel{\text{h}}} \right) \right] 3.80 = \boxed{7.69 \times 10^8 \ \text{J}}$$

(b) The energy extracted from the cold space (outside air) is

$$|Q_c| = |Q_h| - W = |Q_h| - \frac{|Q_h|}{\text{COP}} = |Q_h| \left(1 - \frac{1}{\text{COP}} \right)$$

or $$|Q_c| = \left(7.69 \times 10^8 \ \text{J} \right) \left(1 - \frac{1}{3.80} \right) = \boxed{5.67 \times 10^8 \ \text{J}}$$

***P22.11** $\text{COP} = 3.00 = \dfrac{Q_c}{W}$. Therefore, $W = \dfrac{Q_c}{3.00}$.

The heat removed each minute is

$$\frac{Q_C}{t} = (0.030\ 0\ \text{kg})(4\ 186\ \text{J/kg}\cdot^\circ\text{C})(22.0^\circ\text{C})$$

$$+ (0.030\ 0\ \text{kg})(3.33 \times 10^5\ \text{J/kg})$$

$$+ (0.030\ 0\ \text{kg})(2\ 090\ \text{J/kg}\cdot^\circ\text{C})(20.0^\circ\text{C})$$

$$= 1.40 \times 10^4\ \text{J/min} = 233\ \text{J/s}$$

Thus, the work done per second $= P = \dfrac{233\ \text{J/s}}{3.00} = \boxed{77.8\ \text{W}}$.

P22.12 (a) The coefficient of performance of a heat pump is

$$\text{COP}_{h.p.} = \frac{Q_H}{W} = \frac{Q_H}{Q_H - Q_L}$$

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Because work (energy) is power times time $(W = P\Delta t)$, the equation above may be rearranged to obtain the heat added to the home:

$$Q_H = \text{COP} \cdot W = \text{COP} \cdot P\Delta t$$
$$= (4.20)\left(1.75 \times 10^3 \text{ J/s}\right)(3600 \text{ s}) = \boxed{2.65 \times 10^7 \text{ J}}$$

(b) The coefficient of performance of a refrigerator or air conditioner is

$$\text{COP}_{\text{refr.}} = \frac{Q_L}{W} = \frac{Q_L}{Q_H - Q_L}$$

and can be written in terms of the coefficient of performance of a heat pump because:

$W = Q_H - Q_L$:

$$\text{COP}_{\text{refr.}} = \frac{Q_L}{W} = \frac{Q_H - W}{Q_H - Q_L} = \frac{Q_H}{Q_H - Q_L} + \frac{W}{Q_H - Q_L}$$

Where the first term on the far right is identically the coefficient of performance of the heat pump, and the second term is identically one (because $W = Q_H - Q_L$). Thus,

$$\text{COP}_{\text{refr.}} = \frac{Q_H}{Q_H - Q_L} + \frac{W}{Q_H - Q_L} = \text{COP}_{\text{h.p.}} - 1 = (4.20) - 1$$
$$= \boxed{3.20}$$

P22.13 (a) The energy use by the freezer each day is

$$W = P \cdot \Delta t = \left(457 \ \frac{\cancel{\text{kWh}}}{\cancel{\text{y}}}\right)\left(\frac{3.60 \times 10^6 \text{ J}}{1 \ \cancel{\text{kWh}}}\right)\left(\frac{1 \ \cancel{\text{y}}}{365 \text{ d}}\right) \cdot (1 \text{ d})$$
$$= \boxed{4.51 \times 10^6 \text{ J}}$$

(b) From the definition of the coefficient of performance for a refrigerator, $(\text{COP})_R = |Q_c|/W$, the thermal energy removed from the cold space each day is

$$|Q_c| = (\text{COP})_R \cdot W = 6.30\left(4.51 \times 10^6 \text{ J}\right) = \boxed{2.84 \times 10^7 \text{ J}}$$

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(c) The water must be cooled 20.0°C before it will start to freeze, so the thermal energy that must be removed from mass m of water to freeze it is $|Q_c| = mc_w|\Delta T| + mL_f$. The mass of water that can be frozen each day is then

$$m = \frac{|Q_c|}{c_w|\Delta T| + L_f} = \frac{2.84\times10^7\ \text{J}}{(4186\ \text{J/kg}\cdot°\text{C})(20.0°\text{C}) + 3.33\times10^5\ \text{J/kg}}$$

$$= \boxed{68.1\ \text{kg}}$$

Section 22.3 Reversible and Irreversible Processes

Section 22.4 The Carnot Engine

P22.14 The maximum possible efficiency for a heat engine operating between reservoirs with absolute temperatures of $T_c = 25° + 273 = 298$ K and $T_h = 375° + 273 = 648$ K is the Carnot efficiency:

$$e_c = 1 - \frac{T_c}{T_h} = 1 - \frac{298\ \text{K}}{648\ \text{K}} = \boxed{0.540\ (\text{or } 54.0\%)}$$

P22.15 We use the Carnot expression for maximum possible efficiency, and the definition of efficiency to find the useful output. The engine is a steam turbine in an electric generating station with

$$T_c = 430°\text{C} = 703\ \text{K} \qquad \text{and} \qquad T_h = 1\,870°\text{C} = 2\,143\ \text{K}$$

(a) $$e_C = \frac{\Delta T}{T_h} = \frac{1\,440\ \text{K}}{2\,143\ \text{K}} = 0.672 = \boxed{67.2\%}$$

(b) $$e = W_{\text{eng}}/|Q_h| = 0.420 \qquad \text{and} \qquad |Q_h| = 1.40\times10^5\ \text{J}$$

for one second of operation, so

$$W_{\text{eng}} = 0.420|Q_h| = 5.88\times10^4\ \text{J}$$

and the power is

$$P = \frac{W_{\text{eng}}}{\Delta t} = \frac{5.88\times10^4\ \text{J}}{1\ \text{s}} = \boxed{58.8\ \text{kW}}$$

P22.16 The efficiency of a Carnot engine operating between these temperatures is

$$e_C = 1 - \frac{T_c}{T_h} = 1 - \frac{273\ \text{K}}{293\ \text{K}} = 0.068\ 3 = 6.83\%$$

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Therefore, there is no way that the inventor's engine can have an efficiency of 0.110 = 11.0%.

P22.17 $$e = \frac{W_{\text{eng}}}{|Q_h|} = e_c = 1 - \frac{T_c}{T_h} \quad \rightarrow \quad \frac{W_{\text{eng}} / \Delta t}{|Q_h| / \Delta t} = \frac{P}{|Q_h|/\Delta t} = 1 - \frac{T_c}{T_h}$$

(a) $$|Q_h| = \frac{W_{\text{eng}}}{e} = \frac{P\Delta t}{1 - (T_c / T_h)} = \frac{(1.50 \times 10^5 \text{ W})(3\ 600 \text{ s})}{1 - (293 \text{ K} / 773 \text{ K})} = \boxed{8.70 \times 10^8 \text{ J}}$$

(b) $$|Q_c| = |Q_h| - W_{\text{eng}} = |Q_h| - P\Delta t$$
$$= 8.70 \times 10^8 \text{ J} - (1.50 \times 10^5 \text{ W})(3\ 600 \text{ s})$$
$$= \boxed{3.30 \times 10^8 \text{ J}}$$

P22.18 $$e = \frac{W_{\text{eng}}}{|Q_h|} = e_c = 1 - \frac{T_c}{T_h} \quad \rightarrow \quad \frac{W_{\text{eng}} / \Delta t}{|Q_h| / \Delta t} = \frac{P}{|Q_h|/\Delta t} = 1 - \frac{T_c}{T_h}$$

(a) $$|Q_h| = \frac{W_{\text{eng}}}{e} = \frac{P\Delta t}{1 - (T_c / T_h)} = \boxed{P\Delta t\left(\frac{T_h}{T_h - T_c}\right)}$$

(b) $$|Q_c| = |Q_h| - W_{\text{eng}} = |Q_h| - P\Delta t = P\Delta t\left(\frac{T_h}{T_h - T_c}\right) - P\Delta t$$

$$= P\Delta t\left(\frac{T_h}{T_h - T_c} - 1\right) = P\Delta t\left(\frac{T_h - (T_h - T_c)}{T_h - T_c}\right) = \boxed{P\Delta t\left(\frac{T_c}{T_h - T_c}\right)}$$

P22.19 $$(\text{COP})_{\text{refrig}} = \frac{T_c}{\Delta T} = \frac{270 \text{ K}}{30.0 \text{ K}} = \boxed{9.00}$$

P22.20 (a) For a complete cycle, $\Delta E_{\text{int}} = 0$ and

$$W = |Q_h| - |Q_c| = |Q_c|\left[\frac{|(Q_h)|}{|Q_c|} - 1\right]$$

The text shows that for a Carnot cycle (and only for a reversible cycle), $\frac{|Q_h|}{|Q_c|} = \frac{T_h}{T_c}$. Therefore, $W = \frac{T_h - T_c}{T_c}|Q_c|$.

(b) We have the definition of the coefficient of performance for a refrigerator, $\text{COP} = \frac{|Q_c|}{W}$. Using the result from part (a), this becomes $\text{COP} = \frac{T_c}{T_h - T_c}$.

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P22.21 $$(\text{COP})_{\text{heat pump}} = \frac{|Q_c| + W}{W} = \frac{T_h}{\Delta T} = \frac{295\ \text{K}}{25\ \text{K}} = \boxed{11.8}$$

P22.22 $$(\text{COP})_{\text{Carnot refrig}} = \frac{T_c}{\Delta T} = \frac{4.00\ \text{K}}{289\ \text{K}} = 0.013\ 8 = \frac{|Q_c|}{W}$$

$$\therefore W = \boxed{72.2\ \text{J}}\ \text{per 1 J of energy removed by heat.}$$

P22.23 We wish to evaluate $\text{COP} = |Q_c|/W$ for a refrigerator, which is a Carnot engine run in reverse. For a Carnot engine,

$$\left.\begin{aligned} |Q_h| &= |Q_c| + W \\ e &= \frac{W}{|Q_h|} = \frac{W}{|Q_c| + W} \end{aligned}\right\} \rightarrow \frac{1}{e} = \frac{|Q_c| + W}{W} = \frac{|Q_c|}{W} + 1$$

which gives

$$\text{COP} = \frac{|Q_c|}{W} = \frac{1}{e} - 1$$

Therefore,

$$\text{COP} = \frac{1}{e} - 1 = \frac{1}{0.350} - 1 = \boxed{1.86}.$$

***P22.24** The Carnot summer efficiency is

$$e_{c,s} = 1 - \frac{T_c}{T_h} = 1 - \frac{(273\ \text{K} + 20.0°\text{C})}{(273\ \text{K} + 350°\text{C})} = 0.530$$

And in winter,

$$e_{c,w} = 1 - \frac{283}{623} = 0.546$$

Then the actual winter efficiency is

$$0.320\left(\frac{0.546}{0.530}\right) = \boxed{0.330} \quad \text{or} \quad \boxed{33.0\%}$$

P22.25 (a) The absolute temperature of the cold reservoir is $T_c = 20.0° + 273 = 293$ K. If the Carnot efficiency is to be $e_C = 0.650$, it is necessary that

$$1 - \frac{T_c}{T_h} = 0.650 \quad \text{or} \quad \frac{T_c}{T_h} = 0.350 \quad \text{and} \quad T_h = \frac{T_c}{0.35}$$

Thus,

$$T_h = \frac{293\ \text{K}}{0.35} = 837\ \text{K} \quad \text{or} \quad T_h = 837 - 273 = \boxed{564°\text{C}}$$

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(b) No. A real engine will always have an efficiency *less* than the Carnot efficiency because it operates in an irreversible manner.

P22.26 (a) $e_C = 1 - \frac{T_c}{T_h} = 1 - \frac{350\text{ K}}{500\text{ K}} = \boxed{0.300}$

(b) We differentiate $e_C = 1 - T_c/T_h$ to find

$$\frac{de_c}{dT_h} = 0 - T_c(-1)T_h^{-2} = \frac{T_c}{T_h^2} = \frac{350\text{ K}}{(500\text{ K})^2} = \boxed{1.40\times10^{-3}\text{ K}^{-1}}$$

(c) We differentiate $e_C = 1 - T_c/T_h$ to find

$$\frac{de_c}{dT_c} = 0 - \frac{1}{T_h} = -\frac{1}{500\text{ K}} = \boxed{-2.00\times10^{-3}\text{ K}^{-1}}$$

(d) No. The derivative in part (c) depends only on T_h.

P22.27 Isothermal expansion at $T_h = 523$ K

Isothermal compression at $T_c = 323$ K

Gas absorbs 1 200 J during expansion.

(a) For a Carnot cycle, $e_c = 1 - \frac{T_c}{T_h}$

For any engine, $e = \frac{W_{\text{eng}}}{|Q_h|} = 1 - \frac{|Q_c|}{|Q_h|}$

Therefore, for a Carnot engine, $1 - \frac{T_c}{T_h} = 1 - \frac{|Q_c|}{|Q_h|}$

Then we have

$$|Q_c| = |Q_h|\left(\frac{T_c}{T_h}\right) = (1\ 200\text{ J})\left(\frac{323\text{ K}}{523\text{ K}}\right) = \boxed{741\text{ J}}$$

(b) The work we can calculate as

$$W_{\text{eng}} = |Q_h| - |Q_c| = (1\ 200\text{ J} - 741\text{ J}) = \boxed{459\text{ J}}$$

P22.28 (a) $e_{\max} = 1 - \frac{T_c}{T_h} = 1 - \frac{278\text{ K}}{293\text{ K}} = 5.12\times10^{-2} = \boxed{5.12\%}$

(b) $P = \frac{W_{\text{eng}}}{\Delta t} = 75.0\times10^{6}\text{ J/s}$

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Therefore, $W_{\text{eng}} = (75.0 \times 10^6 \text{ J/s})(3\ 600 \text{ s/h}) = 2.70 \times 10^{11} \text{ J/h}$.

From $e = \dfrac{W_{\text{eng}}}{|Q_h|}$ we find

$$|Q_h| = \frac{W_{\text{eng}}}{e} = \frac{2.70 \times 10^{11} \text{ J/h}}{5.12 \times 10^{-2}} = 5.27 \times 10^{12} \text{ J/h} = \boxed{5.27 \text{ TJ/h}}$$

(c) As fossil-fuel prices rise, this way to use solar energy will become a good buy.

***P22.29** (a) With reservoirs at absolute temperatures of T_c = 80.0°C + 273 = 353 K and T_h = 350°C + 273 = 623 K, the Carnot efficiency is

$$e_C = 1 - \frac{T_c}{T_h} = 1 - \frac{353 \text{ K}}{623 \text{ K}} = \boxed{0.433} \quad \text{(or 43.3\%)}$$

so the maximum power output is

$$P_{\max} = \frac{W_{\text{eng}}}{\Delta t} = \frac{e_C |Q_h|}{\Delta t} = \frac{0.433(21.0 \text{ kJ})}{1.00 \text{ s}} = \boxed{9.10 \text{ kW}}$$

(b) From $e = 1 - \dfrac{|Q_c|}{|Q_h|}$, the energy expelled by heat each cycle is

$$|Q_c| = |Q_h|(1 - e) = (21.0 \text{ kJ})(1 - 0.433) = \boxed{11.9 \text{ kJ}}$$

P22.30 (a) $$e = \frac{W_{\text{eng1}} + W_{\text{eng2}}}{Q_{1h}} = \frac{e_1 Q_{1h} + e_2 Q_{2h}}{Q_{1h}}$$

Now, $Q_{2h} = Q_{1c} = Q_{1h} - W_{\text{eng1}} = Q_{1h} - e_1 Q_{1h}$,

so

$$e = \frac{e_1 Q_{1h} + e_2 (Q_{1h} - e_1 Q_{1h})}{Q_{1h}} = \boxed{e_1 + e_2 - e_1 e_2}$$

(b) $$e = e_1 + e_2 - e_1 e_2 = 1 - \frac{T_i}{T_h} + 1 - \frac{T_c}{T_i} - \left(1 - \frac{T_i}{T_h}\right)\left(1 - \frac{T_c}{T_i}\right)$$

$$= 2 - \frac{T_i}{T_h} - \frac{T_c}{T_i} - 1 + \frac{T_i}{T_h} + \frac{T_c}{T_i} - \frac{T_c}{T_h} = \boxed{1 - \frac{T_c}{T_h}}$$

(c) The combination of reversible engines is itself a reversible engine so it has the Carnot efficiency. No improvement in net efficiency has resulted.

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(d) With $W_{eng2} = W_{eng1}$, $e = \dfrac{W_{eng1} + W_{eng2}}{Q_{1h}} = \dfrac{2W_{eng1}}{Q_{1h}} = 2e_1$

$$1 - \frac{T_c}{T_h} = 2\left(1 - \frac{T_i}{T_h}\right)$$

$$0 - \frac{T_c}{T_h} = 1 - \frac{2T_i}{T_h}$$

$$2T_i = T_h + T_c$$

$$\boxed{T_i = \frac{1}{2}\left(T_h + T_c\right)}$$

(e) $e_1 = e_2 = 1 - \dfrac{T_i}{T_h} = 1 - \dfrac{T_c}{T_i}$

$$T_i^2 = T_c T_h$$

$$\boxed{T_i = \left(T_h T_c\right)^{1/2}}$$

P22.31 (a) In an adiabatic process, $P_f V_f^{\gamma} = P_i V_i^{\gamma}$. Also, $\left(\dfrac{P_f V_f}{T_f}\right)^{\gamma} = \left(\dfrac{P_i V_i}{T_i}\right)^{\gamma}$.

Dividing the second equation by the first yields $T_f = T_i\left(\dfrac{P_f}{P_i}\right)^{(\gamma-1)/\gamma}$.

Since $\gamma = \dfrac{5}{3}$ for argon, $\dfrac{\gamma - 1}{\gamma} = \dfrac{2}{5} = 0.400$ and we have

$$T_f = (1\ 073\ \text{K})\left(\frac{300 \times 10^3\ \text{Pa}}{1.50 \times 10^6\ \text{Pa}}\right)^{0.400} = \boxed{564\ \text{K}}$$

(b) $\Delta E_{int} = nC_V \Delta T = Q - W_{eng} = 0 - W_{eng}$, so $W_{eng} = -nC_V \Delta T$,

and the power output is (suppressing the units of R)

$$P = \frac{W_{eng}}{\Delta t} = \frac{-nC_V \Delta T}{\Delta t}$$

$$= \frac{(-80.0\ \text{kg})(1\ \text{mol}/0.039\ 9\ \text{kg})\left(\dfrac{3}{2}\right)(8.314)(564\ \text{K} - 1\ 073\ \text{K})}{60.0\ \text{s}}$$

$$P = 2.12 \times 10^5\ \text{W} = \boxed{212\ \text{kW}}$$

(c) $e_C = 1 - \dfrac{T_c}{T_h} = 1 - \dfrac{564\ \text{K}}{1\ 073\ \text{K}} = 0.475$ or $\boxed{47.5\%}$

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P22.32 (a) First, consider the adiabatic process $D \to A$:

$$P_D V_D^{\gamma} = P_A V_A^{\gamma} \quad \text{so}$$

$$P_D = P_A \left(\frac{V_A}{V_D} \right)^{\gamma} = (1\ 400\ \text{kPa}) \left(\frac{10.0\ \text{L}}{15.0\ \text{L}} \right)^{5/3} = \boxed{712\ \text{kPa}}$$

Also, $\left(\frac{nRT_D}{V_D} \right) V_D^{\gamma} = \left(\frac{nRT_A}{V_A} \right) V_A^{\gamma}$,

or $T_D = T_A \left(\frac{V_A}{V_D} \right)^{\gamma - 1} = (720\ \text{K}) \left(\frac{10.0}{15.0} \right)^{2/3} = \boxed{549\ \text{K}}$.

Now, consider the isothermal process $C \to D$: $T_C = T_D = \boxed{549\ \text{K}}$

$$P_C = P_D \left(\frac{V_D}{V_C} \right) = \left[P_A \left(\frac{V_A}{V_D} \right)^{\gamma} \right] \left(\frac{V_D}{V_C} \right) = \frac{P_A V_A^{\gamma}}{V_C V_D^{\gamma - 1}}$$

$$P_C = \frac{(1\ 400\ \text{kPa})(10.0\ \text{L})^{5/3}}{(24.0\ \text{L})(15.0\ \text{L})^{2/3}} = \boxed{445\ \text{kPa}}$$

Next, consider the adiabatic process $B \to C$: $P_B V_B^{\gamma} = P_C V_C^{\gamma}$

But, $P_C = \frac{P_A V_A^{\gamma}}{V_C V_D^{\gamma - 1}}$ from above. Also considering the isothermal process, $P_B = P_A \left(\frac{V_A}{V_B} \right)$.

Hence, $P_A \left(\frac{V_A}{V_B} \right) V_B^{\gamma} = \left(\frac{P_A V_A^{\gamma}}{V_C V_D^{\gamma - 1}} \right) V_C^{\gamma}$, which reduces to

$$V_B = \frac{V_A V_C}{V_D} = \frac{(10.0\ \text{L})(24.0\ \text{L})}{15.0\ \text{L}} = \boxed{16.0\ \text{L}}$$

Finally, $P_B = P_A \left(\frac{V_A}{V_B} \right) = (1\ 400\ \text{kPa}) \left(\frac{10.0\ \text{L}}{16.0\ \text{L}} \right) = \boxed{875\ \text{kPa}}$.

State	***P* (kPa)**	***V* (L)**	***T* (K)**
A	1 400	10.0	720
B	875	16.0	720
C	445	24.0	549
D	712	15.0	549

TABLE P22.32(a)

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(b) For the isothermal process $A \to B$: $\Delta E_{\text{int}} = nC_V \Delta T = \boxed{0}$ so

$$Q = -W = nRT \ln\left(\frac{V_B}{V_A}\right)$$

$$= (2.34 \text{ mol})(8.314 \text{ J/mol} \cdot \text{K})(720 \text{ K}) \ln\left(\frac{16.0 \text{ L}}{10.0 \text{ L}}\right)$$

$$= \boxed{+6.58 \text{ kJ}}$$

For the adiabatic process $B \to C$: $Q = \boxed{0}$

$$\Delta E_{\text{int}} = nC_V (T_C - T_B)$$

$$= (2.34 \text{ mol})\left[\frac{3}{2}(8.314 \text{ J/mol} \cdot \text{K})\right](549 \text{ K} - 720 \text{ K})$$

$$= \boxed{-4.98 \text{ kJ}}$$

and $W = -Q + \Delta E_{\text{int}} = 0 + (-4.98 \text{ kJ}) = \boxed{-4.98 \text{ kJ}}$.

For the isothermal process $C \to D$: $\Delta E_{\text{int}} = nC_V \Delta T = \boxed{0}$ and

$$Q = -W = nRT \ln\left(\frac{V_D}{V_C}\right)$$

$$= (2.34 \text{ mol})(8.314 \text{ J/mol} \cdot \text{K})(549 \text{ K}) \ln\left(\frac{15.0 \text{ L}}{24.0 \text{ L}}\right)$$

$$= \boxed{-5.02 \text{ kJ}}$$

Finally, for the adiabatic process $D \to A$: $Q = \boxed{0}$

$$\Delta E_{\text{int}} = nC_V (T_A - T_D)$$

$$= (2.34 \text{ mol})\left[\frac{3}{2}(8.314 \text{ J/mol} \cdot \text{K})\right](720 \text{ K} - 549 \text{ K})$$

$$= \boxed{+4.98 \text{ kJ}}$$

and $W = -Q + \Delta E_{\text{int}} = 0 + 4.98 \text{ kJ} = \boxed{+4.98 \text{ kJ}}$

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Process	Q (kJ)	W (kJ)	ΔE_{int} (kJ)
$A \rightarrow B$	+6.58	–6.58	0
$B \rightarrow C$	0	–4.98	–4.98
$C \rightarrow D$	–5.02	+5.02	0
$D \rightarrow A$	0	+4.98	+4.98
$ABCDA$	+1.56	–1.56	0

TABLE P22.32(b)

The work done *by* the engine is the negative of the work input. The output work W_{eng} is given by the work column in TABLE P22.32(b) with all signs reversed.

(c) $$e = \frac{W_{\text{eng}}}{|Q_h|} = \frac{-W_{ABCD}}{Q_{A\rightarrow B}} = \frac{1.56 \text{ kJ}}{6.58 \text{ kJ}} = 0.237 \quad \text{or} \quad \boxed{23.7\%}$$

(d) $$e_c = 1 - \frac{T_c}{T_h} = 1 - \frac{549 \text{ K}}{720 \text{ K}} = 0.237 \quad \text{or} \quad \boxed{23.7\%}$$

P22.33 (a) "The actual efficiency is two thirds the Carnot efficiency" reads as an equation

$$\frac{W_{\text{eng}}}{|Q_h|} = \frac{W_{\text{eng}}}{|Q_c| + W_{\text{eng}}} = \frac{2}{3}\left(1 - \frac{T_c}{T_h}\right) = \frac{2}{3}\frac{T_h - T_c}{T_h}$$

All the T's represent absolute temperatures. Then

$$\frac{|Q_c| + W_{\text{eng}}}{W_{\text{eng}}} = \frac{1.5\ T_h}{T_h - T_c} \quad \rightarrow \quad \frac{|Q_c|}{W_{\text{eng}}} = \frac{1.5\ T_h}{T_h - T_c} - 1 = \frac{1.5\ T_h - T_h + T_c}{T_h - T_c}$$

$$|Q_c| = W_{\text{eng}}\frac{0.5\ T_h + T_c}{T_h - T_c} \rightarrow \frac{|Q_c|}{\Delta t} = \frac{W_{\text{eng}}}{\Delta t}\frac{0.5\ T_h + T_c}{T_h - T_c}$$

$\boxed{\dfrac{Q_c}{\Delta t} = 1.40\left(\dfrac{0.5T_h + 383}{T_h - 383}\right)\text{, where } Q_c/\Delta t \text{ is in megawatts and } T \text{ is in kelvins.}}$

(b) $\boxed{\text{The exhaust power decreases as the firebox temperature increases.}}$

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(c) $$\frac{|Q_c|}{\Delta t} = (1.40\text{ MW})\left(\frac{0.5\,T_h + 383\text{ K}}{T_h - 383\text{ K}}\right)$$

$$= (1.40\text{ MW})\left(\frac{0.5(1\,073\text{ K}) + 383\text{ K}}{1\,073\text{ K} - 383\text{ K}}\right) = \boxed{1.87\text{ MW}}$$

(d) We require

$$\frac{|Q_c|}{\Delta t} = \frac{1}{2}(1.87\text{ MW}) = (1.40\text{ MW})\left(\frac{0.5\,T_h + 383\text{ K}}{T_h - 383\text{ K}}\right)$$

$$\frac{0.5\,T_h + 383\text{ K}}{T_h - 383\text{ K}} = 0.666$$

$$0.5\,T_h + 383\text{ K} = 0.666T_h - 255\text{ K}$$

$$T_h = 638\text{ K}/0.166 = \boxed{3.84 \times 10^3\text{ K}}$$

(e) The minimum possible heat exhaust power is approached as the firebox temperature goes to infinity, and it is $|Q_c|/\Delta t =$ 1.40 MW(0.5/1) = 0.700 MW. The heat exhaust power cannot be as small as (1/4)(1.87 MW) = 0.466 MW. So no answer exists. The energy exhaust cannot be that small.

P22.34 We determine the power required from

$$\frac{|Q_c|}{W} = \text{COP}_c\ (\text{refrigerator}) = \frac{T_c}{T_h - T_c} = \frac{|Q_c|/\Delta t}{W/\Delta t}$$

$$\frac{0.150\text{ W}}{W/\Delta t} = \frac{260\text{ K}}{40.0\text{ K}}$$

$$P = \frac{W}{\Delta t} = (0.150\text{ W})\left(\frac{40.0\text{ K}}{260\text{ K}}\right) = \boxed{23.1\text{ mW}}$$

P22.35 The coefficient of performance of the device is

$$\text{COP} = 0.100\ \text{COP}_{\text{Carnot cycle}}$$

or

$$\frac{|Q_h|}{W} = 0.100\left(\frac{|Q_h|}{W}\right)_{\text{Carnot cycle}} = 0.100\left(\frac{1}{\text{Carnot efficiency}}\right)$$

$$\frac{|Q_h|}{W} = 0.100\left(\frac{T_h}{T_h - T_c}\right) = 0.100\left(\frac{293\text{ K}}{293\text{ K} - 268\text{ K}}\right) = \boxed{1.17}$$

Thus, 1.17 joules of energy enter the room by heat for each joule of work done.

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Section 22.5 Gasoline and Diesel Engines

P22.36 Compression ratio = 6.00, $\gamma = 1.40$

(a) Efficiency of an Otto engine: $e = 1 - \left(\frac{V_2}{V_1}\right)^{\gamma - 1}$

$$e = 1 - \left(\frac{1}{6.00}\right)^{0.400} = \boxed{51.2\%}$$

(b) If actual efficiency $e' = 15.0\%$, the fraction of fuel wasted is (assuming complete combustion of the air-fuel mixture) $e - e' = \boxed{36.2\%}$.

P22.37 (a) For adiabatic expansion, $P_i V_i^{\gamma} = P_f V_f^{\gamma}$. Therefore,

$$P_f = P_i\left(\frac{V_i}{V_f}\right)^{\gamma} = (3.00\times 10^6\ \text{Pa})\left(\frac{50.0\ \text{cm}^3}{300\ \text{cm}^3}\right)^{1.40}$$

$$= \boxed{2.44\times 10^5\ \text{Pa}}$$

(b) Since $Q = 0$, we have $W_{\text{eng}} = Q - \Delta E = -nC_V\Delta T = -nC_V(T_f - T_i)$.

From $\gamma = \frac{C_P}{C_V} = \frac{C_V + R}{C_V}$, we get $(\gamma - 1)C_V = R$, so that

$$C_V = \frac{R}{1.40 - 1} = 2.50\ R$$

The work done by the gas in expanding is then

$$W_{\text{eng}} = n(2.50\ R)(T_i - T_f) = 2.50P_iV_i - 2.50P_fV_f$$

$$= 2.50\left[\left(3.00\times 10^6\ \text{Pa}\right)\left(50.0\times 10^{-6}\ \text{m}^3\right) - \left(2.44\times 10^5\ \text{Pa}\right)\left(300\times 10^{-6}\ \text{m}^3\right)\right]$$

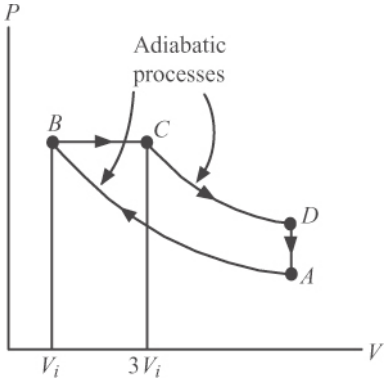
$$= \boxed{192\ \text{J}}$$

P22.38 The energy transferred by heat over the paths *CD* and *BA* is zero since they are adiabatic.

Over path *BC*: $Q_{BC} = nC_P\,(T_C - T_B) > 0$

Over path *DA*: $Q_{DA} = nC_V\,(T_A - T_D) < 0$

Therefore, $|Q_c| = |Q_{DA}|$ and $Q_h = Q_{BC}$.



ANS. FIG. P22.38

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The efficiency is then

$$e = 1 - \frac{|Q_c|}{Q_h} = 1 - \frac{(T_D - T_A)C_V}{(T_C - T_B)C_P} = 1 - \frac{1}{\gamma}\left(\frac{T_D - T_A}{T_C - T_B}\right)$$

Section 22.6 Entropy

P22.39 Each marble is returned to the bag before the next is drawn, so the probability of drawing a red one is the same as drawing a green one.

(a)

Result	Possible Combinations	Total
All red	RRR	1
2R, 1G	RRG, RGR, GRR	3
1R, 2G	RGG, GRG, GGR	3
All green	GGG	1

TABLE P22.39(a)

(b)

Result	Possible Combinations	Total
All red	RRRRR	1
4R, 1G	RRRRG, RRRGR, RRGRR, RGRRR, GRRRR	5
3R, 2G	RRRGG, RRGRG, RGRRG, GRRRG, RRGGR, RGRGR, GRRGR, RGGRR, GRGRR, GGRRR	10
2R, 3G	GGGRR, GGRGR, GRGGR, RGGGR, GGRRG, GRGRG, RGGRG, GRRGG, RGRGG, RRGGG	10
1R, 4G	RGGGG, GRGGG, GGRGG, GGGRG, GGGGR	5
All green	GGGGG	1

TABLE P22.39(b)

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P22.40 (a) The table is shown in TABLE P22.40 below.

(b) On the basis of the table, the most probable recorded result of a toss is $\boxed{\text{2 heads and 2 tails}}$.

Result	Possible Combinations	Total
All heads	HHHH	1
3H, 1T	THHH, HTHH, HHTH, HHHT	4
2H, 2T	TTHH, THTH, THHT, HTTH, HTHT, HHTT	6
1H, 3T	HTTT, THTT, TTHT, TTTH	4
All tails	TTTT	1

TABLE P22.40

P22.41 (a) A 12 can only be obtained $\boxed{\text{one}}$ way, as 6 + 6.

(b) A 7 can be obtained $\boxed{\text{six}}$ ways: 6 + 1, 5 + 2, 4 + 3, 3 + 4, 2 + 5, and 1 + 6.

Section 22.7 Changes in Entropy for Thermodynamic Systems

Section 22.8 Entropy and the Second Law

P22.42 For a freezing process,

$$\Delta S = \frac{\Delta Q}{T} = \frac{-(0.500\text{ kg})(3.33\times10^5\text{ J/kg})}{273\text{ K}} = \boxed{-610\text{ J/K}}$$

P22.43 The hot water has negative energy input by heat, given by $Q = mc\Delta T$. The surrounding room has positive energy input of this same number of joules, which we can write as $Q_{\text{room}} = (mc|\Delta T|)_{\text{water}}$. Imagine the room absorbing this energy reversibly by heat, from a stove at 20.001°C. Then its entropy increase is Q_{room}/T:

$$\Delta S = \frac{Q_r}{T} = \frac{mc_w|\Delta T|}{T} = \frac{(0.125\text{ kg})(4186\text{ J/kg}\cdot\text{°C})(80\text{°C})}{293\text{ K}}$$
$$= \boxed{143\text{ J/K}}$$

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***P22.44** $c_{\text{iron}} = 448\ \text{J/kg}\cdot°\text{C};\ c_{\text{water}} = 4\,186\ \text{J/kg}\cdot°\text{C}$

From $Q_{\text{cold}} = -Q_{\text{hot}}$:

$$(4.00\ \text{kg})(4\,186\ \text{J/kg}\cdot°\text{C})\left(T_f - 10.0°\text{C}\right) = -(1.00\ \text{kg})(448\ \text{J/kg}\cdot°\text{C})\left(T_f - 900°\text{C}\right)$$

which yields $T_f = 33.2°\text{C} = 306.3\ \text{K}$. Then,

$$\Delta S = \int_{283\ \text{K}}^{306.3\ \text{K}} \frac{c_{\text{water}} m_{\text{water}} dT}{T} + \int_{1\,173\ \text{K}}^{306.3\ \text{K}} \frac{c_{\text{iron}} m_{\text{iron}} dT}{T}$$

$$\Delta S = c_{\text{water}} m_{\text{water}} \ln\left(\frac{306.3\ \text{K}}{283\ \text{K}}\right) + c_{\text{iron}} m_{\text{iron}} \ln\left(\frac{306.3\ \text{K}}{1\,173\ \text{K}}\right)$$

$$\Delta S = (4\,186\ \text{J/kg}\cdot\text{K})(4.00\ \text{kg})(0.078\,7) + (448\ \text{J/kg}\cdot\text{K})(1.00\ \text{kg})(-1.34)$$

$$\Delta S = \boxed{717\ \text{J/K}}$$

***P22.45** The car ends up in the same thermodynamic state as it started, so it undergoes zero changes in entropy. The original kinetic energy of the car is transferred by heat to the surrounding air, adding to the internal energy of the air. Its change in entropy is

$$\Delta S = \frac{\frac{1}{2}mv^2}{T} = \frac{\frac{1}{2}(1\,500\ \text{kg})(20.0\ \text{m/s})^2}{293\ \text{K}} = \boxed{1.02\ \text{kJ/K}}$$

P22.46 The total momentum before collision is zero, so the combined mass must be at rest after the collision. The energy dissipated by heat equals the total initial kinetic energy,

$$Q = 2\left(\frac{1}{2}mv^2\right) = (2000\ \text{kg})(20.0\ \text{m/s})^2 = 8.00\times10^5\ \text{J} = 800\ \text{kJ}$$

With the environment at an absolute temperature of $T = 23 + 273 = 296$ K, the change in entropy is

$$\Delta S = \frac{\Delta Q_r}{T} = \frac{800\ \text{kJ}}{296\ \text{K}} = \boxed{2.70\ \text{kJ/K}}$$

P22.47 The potential energy lost by the log is eventually transferred by heat into thermal energy of the environment, so $Q = mgh$, and the change in entropy is

$$\Delta S = \frac{Q}{T} = \frac{mgh}{T} = \frac{(70.0\ \text{kg})(9.80\ \text{m/s}^2)(25.0\ \text{m})}{300\ \text{K}} = \boxed{57.2\ \text{J/K}}$$

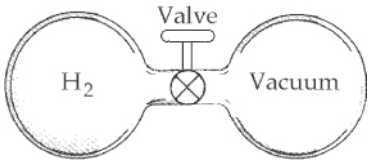
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P22.48 (a) This is a free expansion process. From Equation 22.17,

$$\Delta S = nR\ln\left(\frac{V_f}{V_i}\right) = (1.00\ \text{mol})(8.314\ \text{J/mol}\cdot\text{K})\ln\left(\frac{2}{1}\right)$$

$$= \boxed{5.76\ \text{J/K}}$$

(b) The gas is expanding into an evacuated region. Therefore, $W = 0$. It expands so fast that energy has no time to flow by heat: $Q = 0$. But $\Delta E_{\text{int}} = Q + W$, so in this case $\Delta E_{\text{int}} = 0$. For an ideal gas, the internal energy is a function of the temperature and no other variables, so with $\Delta E_{\text{int}} = 0$, there is $\boxed{\text{no change in temperature}}$.



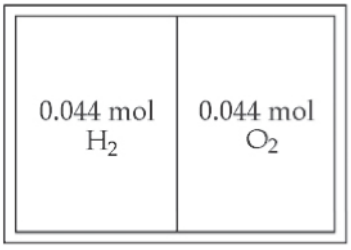
ANS. FIG. P22.48

P22.49 Each gas expands into the other half of the container as though the other gas were not there; therefore, consider each gas to undergo a free expansion process in which its volume doubles. From Equation 22.17, the entropy change is twice that for a single gas:

$$\Delta S = 2\left[nR\ln\left(\frac{V_f}{V_i}\right)\right]$$

$$= 2\left[(0.044\ 0)(8.314\ \text{J/mol}\cdot\text{K})(\ln 2)\right]$$

$$= \boxed{0.507\ \text{J/K}}$$



ANS. FIG. P22.49

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***P22.50** We take data from Tables 20.1 and 20.2, and we assume a constant specific heat for each phase. As the ice is warmed from –12.0°C to 0°C, its entropy increases by

$$\Delta S = \int_i^f \frac{dQ}{T} = \int_{261\text{ K}}^{273\text{ K}} \frac{mc_{\text{ice}}\,dT}{T} = mc_{\text{ice}} \int_{261\text{ K}}^{273\text{ K}} T^{-1}\,dT = mc_{\text{ice}} \ln T\Big|_{261\text{ K}}^{273\text{ K}}$$

$$\Delta S = (0.027\,9\text{ kg})(2\,090\text{ J/kg}\cdot°\text{C})(\ln 273\text{ K} - \ln 261\text{ K})$$

$$= (0.027\,9\text{ kg})(2\,090\text{ J/kg}\cdot°\text{C})\left[\ln\left(\frac{273\text{ K}}{261\text{ K}}\right)\right]$$

$$\Delta S = 2.62\text{ J/K}$$

As the ice melts its entropy change is

$$\Delta S = \frac{Q}{T} = \frac{mL_f}{T} = \frac{(0.027\,9\text{ kg})(3.33\times10^5\text{ J/kg})}{273\text{ K}} = 34.0\text{ J/K}$$

As liquid water warms from 273 K to 373 K,

$$\Delta S = \int_i^f \frac{mc_{\text{liquid}}\,dT}{T} = mc_{\text{liquid}} \ln\left(\frac{T_f}{T_i}\right)$$

$$= (0.027\,9\text{ kg})(4\,186\text{ J/kg}\cdot°\text{C})\ln\left(\frac{373\text{ K}}{273\text{ K}}\right) = 36.5\text{ J/K}$$

As the water boils and the steam warms,

$$\Delta S = \frac{mL_v}{T} + mc_{\text{steam}} \ln\left(\frac{T_f}{T_i}\right)$$

$$\Delta S = \frac{(0.027\,9\text{ kg})(2.26\times10^6\text{ J/kg})}{373\text{ K}} + (0.027\,9\text{ kg})(2\,010\text{ J/kg}\cdot°\text{C})\ln\left(\frac{388\text{ K}}{373\text{ K}}\right)$$

$$= 169\text{ J/K} + 2.21\text{ J/K}$$

The total entropy change is

$$\Delta S_{\text{tot}} = (2.62 + 34.0 + 36.5 + 169 + 2.21)\text{ J/K} = \boxed{244\text{ J/K}}$$

For steam at constant pressure, the molar specific heat in Table 20.1 implies a specific heat of $(35.4\text{ J/mol}\cdot\text{K})\left(\frac{1\text{ mol}}{0.018\text{ kg}}\right) = 1\,970\text{ J/kg}\cdot\text{K}$, nearly agreeing with 2 010 J/kg · K.

***P22.51** The change in entropy is given by

$$\Delta S = \int_i^f \frac{dQ}{T} = \int_{T_i}^{T_f} mc\,\frac{dT}{T}$$

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Here T means the absolute temperature. We would ordinarily think of dT as the change in the Celsius temperature, but one Celsius degree of temperature change is the same size as one kelvin of change, so dT is also the change in absolute T.

$$\Delta S = mc\ln T\Big|_{T_i}^{T_f} = mc\ln\left(\frac{T_f}{T_i}\right)$$

$$= (0.250\text{ kg})(4\,186\text{ J/kg}\cdot\text{K})\ln\left(\frac{353\text{ K}}{293\text{ K}}\right) = \boxed{195\text{ J/K}}$$

P22.52 Sitting here writing, I convert chemical energy from molecules in food, into internal energy that leaves my body by heat into the room-temperature surroundings. My rate of energy output is equal to my metabolic rate,

$$2\,500\text{ kcal/d} = \left(\frac{2\,500\times10^3\text{ cal}}{86\,400\text{ s}}\right)\left(\frac{4.186\text{ J}}{1\text{ cal}}\right) = 120\text{ W}$$

My body is in steady state, changing little in entropy, as the environment increases in entropy at the rate

$$\frac{\Delta S}{\Delta t} = \frac{Q/T}{\Delta t} = \frac{Q/\Delta t}{T} = \frac{120\text{ W}}{293\text{ K}} = 0.4\text{ W/K} \sim \boxed{1\text{ W/K}}$$

When using powerful appliances or an automobile, my personal contribution to entropy production is much greater than the above estimate, based only on metabolism.

P22.53 The change in entropy of a reservoir is $\Delta S = Q_r/T$, where Q_r is the energy absorbed ($Q_r > 0$) or expelled ($Q_r < 0$) by the reservoir, and T is the absolute temperature of the reservoir.

(a) For the hot reservoir: $\Delta S_h = \dfrac{-2.50\times10^3\text{ J}}{725\text{ K}} = \boxed{-3.45\text{ J/K}}$

(b) For the cold reservoir: $\Delta S_c = \dfrac{+2.50\times10^3\text{ J}}{310\text{ K}} = \boxed{+8.06\text{ J/K}}$

(c) For the Universe:

$$\Delta S_U = \Delta S_h + \Delta S_c = -3.45\text{ J/K} + 8.06\text{ J/K} = \boxed{+4.61\text{ J/K}}$$

P22.54 The change in entropy of a reservoir is $\Delta S = Q_r/T$, where Q_r is the energy absorbed ($Q_r > 0$) or expelled ($Q_r < 0$) by the reservoir, and T is the absolute temperature of the reservoir.

(a) Energy is transferred from the hot reservoir by heat: $|Q_h| = -Q$,

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and $\boxed{\Delta S_h = -\frac{Q}{T_h}}$.

(b) Energy is transferred to the cold reservoir by heat: $|Q_c| = +Q$, and $\boxed{\Delta S_c = \frac{Q}{T_c}}$.

(c) For the Universe, $\Delta S_U = \Delta S_h + \Delta S_c = \boxed{Q\left(\frac{1}{T_c} - \frac{1}{T_h}\right)}$.

P22.55 $\Delta S = \frac{Q_2}{T_2} - \frac{Q_1}{T_1} = \left(\frac{1\ 000\ \text{J}}{290\ \text{K}} - \frac{1\ 000\ \text{J}}{5\ 800\ \text{K}}\right) = \boxed{3.28\ \text{J/K}}$

***P22.56** Define T_1 = Temp Cream = 5.00°C = 278 K.

Define T_2 = Temp Coffee = 60.0°C = 333 K.

The final temperature of the mixture is

$$T_f = \frac{(20.0\ \text{g})T_1 + (200\ \text{g})T_2}{220\ \text{g}} = 55.0°\text{C} = 328\ \text{K}$$

The entropy change due to this mixing is

$$\Delta S = (20.0\ \text{g})\int_{T_1}^{T_f} \frac{c_V dT}{T} + (200\ \text{g})\int_{T_2}^{T_f} \frac{c_V dT}{T}$$

$$= (84.0\ \text{J/K})\ln\left(\frac{T_f}{T_1}\right) + (840\ \text{J/K})\ln\left(\frac{T_f}{T_2}\right)$$

$$= (84.0\ \text{K/J})\ln\left(\frac{328\ \text{K}}{278\ \text{K}}\right) + (840\ \text{J/K})\ln\left(\frac{328\ \text{K}}{333\ \text{K}}\right)$$

$$\Delta S = \boxed{+1.18\ \text{J/K}}$$

***P22.57** We first determine the energy that must be extracted from tap water at 10.0°C to produce ice at –20.0°C:

$$|Q_c| = mc\Delta T + mL + mc\Delta T$$

$$|Q_c| = (0.500\ \text{kg})(4\ 186\ \text{J/kg}\cdot°\text{C})(10.0°\text{C}) + (0.500\ \text{kg})(3.33\times10^5\ \text{J/kg}) + (0.500\ \text{kg})(2\ 090\ \text{J/kg}\cdot°\text{C})(20.0°\text{C})$$

$$= 2.08\times10^5\ \text{J}$$

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The work required to accomplish this is then found from

$$\frac{|Q_c|}{W} = \text{COP}_c\,(\text{refrigerator}) = \frac{T_c}{T_h - T_c}$$

or

$$W = \frac{|Q_c|(T_h - T_c)}{T_c} = \frac{(2.08\times10^5\ \text{J})[20.0°\text{C} - (-20.0°\text{C})]}{273\ \text{K} - 20.0°\text{C}}$$

$$= \boxed{32.9\ \text{kJ}}$$

P22.58 We are given $T_c = 273$ K.

(a) For steam at 100°C, $T_h = 373$ K and

$$e = 1 - \frac{T_c}{T_h} = 1 - \frac{273\ \text{K}}{373\ \text{K}} = \boxed{0.268}$$

(b) For superheated steam at 200°C, $T_h = 473$ K and

$$e = 1 - \frac{T_c}{T_h} = 1 - \frac{273\ \text{K}}{473\ \text{K}} = \boxed{0.423}$$

P22.59 $|Q_h| = 3W,$ and for an engine, $|Q_h| = W + |Q_c| = 3W \quad \rightarrow \quad |Q_c| = 2W.$

(a) $e = \dfrac{W}{|Q_h|} = \dfrac{W}{3W} = \boxed{\dfrac{1}{3}}$

(b) $\dfrac{|Q_c|}{|Q_h|} = \dfrac{2W}{3W} = \boxed{\dfrac{2}{3}}$

P22.60 The conversion of gravitational potential energy into kinetic energy as the water falls is reversible. But the subsequent conversion into internal energy is not. We imagine arriving at the same final state by adding energy by heat, in amount mgy, to the water from a stove at a temperature infinitesimally above 20.0°C. Then,

$$\frac{\Delta S}{\Delta t} = \int \frac{dQ_r/\Delta t}{T} = \frac{Q/\Delta t}{T} = \frac{mgy/\Delta t}{T}$$

$$= \frac{(5\ 000\ \text{m}^3/\text{s})(1\ 000\ \text{kg/m}^3)(9.80\ \text{m/s}^2)(50.0\ \text{m})}{293\ \text{K}}$$

$$= \boxed{8.36\times10^6\ \text{J/K}\cdot\text{s}}$$

P22.61 The maximum (Carnot) or ideal efficiency is

$$e_{\text{ideal}} = 1 - \frac{T_c}{T_h}$$

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(where we note that the temperatures must be given in kelvin.) Thus,

$$e_{\text{ideal}} = 1 - \frac{T_c}{T_h} = 1 - \frac{(185°C + 273°)}{(545°C + 273°)} = 1 - \frac{458.15K}{818.15K}$$

$$= \boxed{0.440 = 44.0\%}$$

P22.62 (a) $$\left(10.0\ \frac{\text{Btu/h}}{\text{W}}\right)\left(\frac{1\ 055\ \text{J}}{1\ \text{Btu}}\right)\left(\frac{1\ \text{h}}{3\ 600\ \text{s}}\right)\left(\frac{1\ \text{W}}{1\ \text{J/s}}\right) = \boxed{2.93}$$

(b) The energy extracted by heat from the cold side divided by required work input is by definition the coefficient of performance for a refrigerator: $\boxed{(\text{COP})_{\text{refrigerator}}}$

(c) With an EER of 5,

$$5\ \frac{\text{Btu}}{\text{h}\cdot\text{W}} = \frac{10\ 000\ \text{Btu/h}}{P}$$

which gives

$$P = \frac{10\ 000\ \text{Btu/h}}{5\ \text{Btu/h}\cdot\text{W}} = 2\ 000\ \text{W} = 2.00\ \text{kW}$$

Energy purchased is $P\Delta t = (2.00\ \text{kW})(1\ 500\ \text{h}) = 3.00 \times 10^3\ \text{kWh}$.

$$\text{Cost} = \left(3.00 \times 10^3\ \text{kWh}\right)\left(0.170\ \$/\text{kWh}\right) = \$510:$$

$\boxed{\text{With EER 5, \$510}}$

With EER 10,

$$10\ \frac{\text{Btu}}{\text{h}\cdot\text{W}} = \frac{10\ 000\ \text{Btu/h}}{P}$$

$$\rightarrow P = \frac{10\ 000\ \text{Btu/h}}{10\ \text{Btu/h}\cdot\text{W}} = 1\ 000\ \text{W} = 1.00\ \text{kW}$$

Energy purchased is $P\Delta t = (1.00\ \text{kW})(1\ 500\ \text{h}) = 1.50 \times 10^3\ \text{kWh}$

$$\text{Cost} = \left(1.50 \times 10^3\ \text{kWh}\right)\left(0.170\ \$/\text{kWh}\right) = \$255:$$

$\boxed{\text{With EER 10, \$255}}$

Thus, the cost for air conditioning is half as much for an air conditioner with EER 10 compared with an air conditioner with EER 5.

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P22.63 (a) $P_{\text{electric}} = \dfrac{H_{ET}}{\Delta t}$ so if all the electric energy is converted into internal energy, the steady-state condition of the house is described by $H_{ET} = |Q|$.

Therefore, $P_{\text{electric}} = \dfrac{Q}{\Delta t} = \boxed{5.00 \text{ kW}}$.

(b) For a heat pump, $(\text{COP})_{\text{Carnot}} = \dfrac{T_h}{\Delta T} = \dfrac{295 \text{ K}}{27.0 \text{ K}} = 10.93$

Actual $\text{COP} = 0.6(10.93) = 6.56 = \dfrac{|Q_h|}{W} = \dfrac{|Q_h|/\Delta t}{W/\Delta t}$

Therefore, to bring 5 000 W of energy into the house only requires input power

$$P_{\text{heat pump}} = \frac{W}{\Delta t} = \frac{|Q_h|/\Delta t}{\text{COP}} = \frac{5\ 000 \text{ W}}{6.56} = \boxed{763 \text{ W}}$$

***P22.64** (a) The energy transferred to the gas by heat is

$$Q = mc\Delta T = (1.00 \text{ mol})(20.79 \text{ J/mol} \cdot \text{K})(120 \text{ K})$$
$$= 2.49 \times 10^3 \text{ J} = \boxed{2.49 \text{ kJ}}$$

(b) Treating the neon as a monatomic ideal gas, Equation 21.25 gives the change in internal energy as $\Delta U = \dfrac{3}{2} nR\Delta T$, or

$$\Delta U = \frac{3}{2}(1.00 \text{ mol})(8.314 \text{ J/mol} \cdot \text{K})(120 \text{ K})$$
$$= 1.50 \times 10^3 \text{ J} = \boxed{1.50 \text{ kJ}}$$

(c) From the first law, the work done on the gas is

$$W = \Delta U - Q = 1.50 \times 10^3 \text{ J} - 2.49 \times 10^3 \text{ J} = \boxed{-990 \text{ J}}$$

P22.65 Energy transfer by heat for infinitesimal temperature change dT is $dQ = nCdT$, where C is the molar specific heat for either constant volume ($C_V = 5R/2$) or pressure ($C_P = 7R/2$) for air, a diatomic gas. The corresponding entropy change is

$$dS = \frac{dQ_r}{T} = \frac{nCdT}{T} \quad \rightarrow \quad \Delta S = \int_{T_i}^{T_f} \frac{nCdT}{T} = nC \ln \frac{T_f}{T_i},$$

with $T_i = 25.0 + 273 = 298 \text{ K}$ and $T_f = -18.0 + 273 = 255 \text{ K}$.

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(a) $$\Delta S = nC_V \ln\frac{T_f}{T_i} = n\frac{5}{2}R\ln\left(\frac{255\text{ K}}{298\text{ K}}\right) = \boxed{-0.390nR}$$

(b) $$\Delta S = nC_P \ln\frac{T_f}{T_i} = n\frac{7}{2}R\ln\left(\frac{255\text{ K}}{298\text{ K}}\right) = \boxed{-0.545nR}$$

P22.66 (a) The coefficient of performance of an air conditioner is defined as

$$(\text{COP})_{\text{ac}} = \frac{|Q_c|}{W} = \frac{|Q_c|}{|Q_h| - |Q_c|} = \frac{1}{|Q_h|/|Q_c| - 1}$$

But when a device operates on the Carnot cycle, $|Q_h|/|Q_c| = T_h/T_c$. Thus, the coefficient of performance for a Carnot heat pump would be

$$(\text{COP})_{\text{ac}} = \frac{1}{T_h/T_c - 1} = \boxed{\frac{T_c}{T_h - T_c}}$$

(b) From the result of part (a) above, we observe that the COP of a Carnot air conditioner would increase if the temperature difference $T_h - T_c$ becomes $\boxed{\text{smaller}}$.

(c) If $T_c = 20° + 273 = 293\text{ K}$ and $T_h = 40° + 273 = 313\text{ K}$, the COP of a Carnot heat pump would be

$$(\text{COP})_{\text{ac},C} = \frac{T_c}{T_h - T_c} = \frac{293\text{ K}}{313\text{ K} - 293\text{ K}} = \boxed{14.6}$$

P22.67 (a) For the constant volume process AB,

$$Q_{\text{AB}} = \Delta E_{\text{int},\,AB} = \frac{3}{2}nR\Delta T = \frac{3}{2}nR(3T_i - T_i) = \boxed{3nRT_i}$$

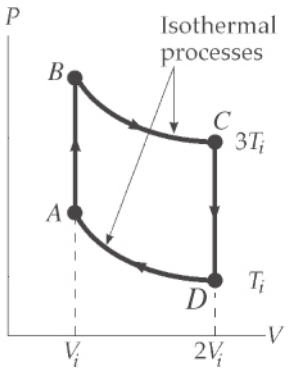
(b) For an isothermal process, $Q = nRT\ln\left(\frac{V_2}{V_1}\right)$.

Therefore, for process BC, $Q_{BC} = \boxed{3nRT_i \ln 2}$.

(c) For the constant volume process CD,

$$Q_{CD} = \Delta E_{\text{int},\,CD} = \frac{3}{2}nR\Delta T = \frac{3}{2}nR(T_i - 3T_i) = \boxed{-3nRT_i}$$

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ANS. FIG. P22.67

(d) For an isothermal process DA, $Q_{DA} = nRT_i \ln\frac{1}{2} = \boxed{-nRT_i \ln 2}$.

(e) $Q_h = Q_{AB} + Q_{BC} = 3nRT_i + 3nRT_i \ln 2 = \boxed{3nRT_i(1+\ln 2)}$

(f) Since the change in temperature for the complete cycle is zero,

$$\Delta E_{\text{int}} = 0 \quad \text{and} \quad W_{\text{eng}} = Q$$

and work done by the engine is

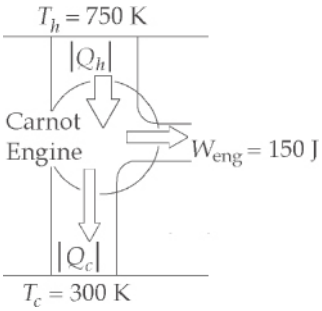
$$W = Q = Q_{\text{AB}} + Q_{\text{BC}} + Q_{\text{CD}} + Q_{\text{DA}}$$
$$= 3nRT_i + 3nRT_i \ln 2 - 3nRT_i - nRT_i \ln 2$$
$$W = \boxed{2nRT_i \ln 2}$$

(g) The efficiency is

$$e_{\text{c}} = \frac{W_{\text{eng}}}{|Q_h|} = \frac{Q}{|Q_h|} = \frac{2\ln 2}{3(1+\ln 2)} = \boxed{0.273}$$

P22.68 For the Carnot engine,

$$e_{\text{c}} = 1 - \frac{T_c}{T_h} = 1 - \frac{300\text{ K}}{750\text{ K}} = 0.600$$



ANS. FIG. P22.68

Also, $e_{\text{c}} = \dfrac{W_{\text{eng}}}{|Q_h|}$,

so $|Q_h| = \dfrac{W_{\text{eng}}}{e_{\text{c}}} = \dfrac{150\text{ J}}{0.600} = 250\text{ J}$

and $|Q_c| = |Q_h| - W_{\text{eng}} = 250\text{ J} - 150\text{ J} = 100\text{ J}.$

(a) $|Q_h| = \dfrac{W_{\text{eng}}}{e_{\text{S}}} = \dfrac{150\text{ J}}{0.700} = \boxed{214\text{ J}}$

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$$|Q_c| = |Q_h| - W_{\text{eng}} = \frac{W_{\text{eng}}}{e_S} - 150\text{ J} = \boxed{64.3\text{ J}}$$

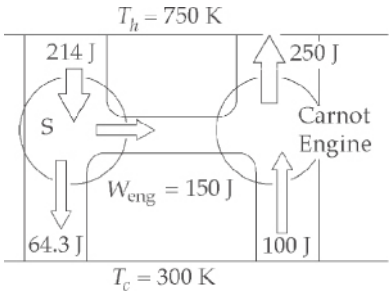
(b) When engine S delivers 150 J of work to the Carnot engine, the Carnot engine transfers 250 J to the firebox while engine S takes 214 J from the firebox:

$$|Q_{h,\text{net}}| = -\frac{W_{\text{eng}}}{e_S} + 250\text{ J} = \boxed{35.7\text{ J}}$$

and the Carnot engine removes 100 J from the environment while engine S returns 64.3 J:

$$|Q_{c,\text{net}}| = |64.3\text{ J} - 100\text{ J}| = \boxed{35.7\text{ J}}$$

The total energy the firebox puts out equals the total energy transferred to the environment.



ANS. FIG. P22.68(a–b)

(c) The net flow of energy by heat from the cold to the hot reservoir without work input is impossible.

(d) For engine S: $|Q_{c,S}| = |Q_{h,S}| - W_{\text{eng S}} = \dfrac{W_{\text{eng S}}}{e_S} - W_{\text{eng S}}$

so work output is $W_{\text{eng S}} = \dfrac{|Q_{c,S}|}{\frac{1}{e_S} - 1} = \dfrac{100\text{ J}}{\frac{1}{0.700} - 1} = \boxed{233\text{ J}}$

and energy input to engine S is

$$|Q_{h,S}| = |Q_{c,S}| + W_{\text{eng S}} = 233\text{ J} + 100\text{ J} = \boxed{333\text{ J}}$$

(e) Engine S contributes 150 J out of 233 J to running the Carnot engine:

$$|Q_{h,\text{net}}| = |Q_{h,S}| - 250\text{ J} = 333\text{ J} - 250\text{ J} = \boxed{83.3\text{ J}}$$

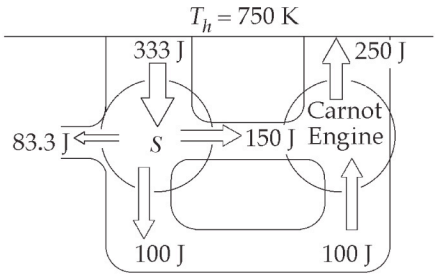
This is the net energy lost by the firebox.

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(f) The remaining work output is

$$W_{\text{net}} = W_{\text{eng S}} - 250\text{ J} = 233\text{ J} - 150\text{ J} = \boxed{83.3\text{ J}}$$

(g) $\left|Q_{c,\text{net}}\right| = \boxed{0}$



ANS. FIG. P22.68(e–g)

(h) The output of 83.3 J of energy from the heat engine by work in a cyclic process without any exhaust by heat is impossible.

(i) Both engines operate in cycles, so $\Delta S_{\text{S}} = \Delta S_{\text{Carnot}} = 0$.

For the reservoirs, $\Delta S_h = -\dfrac{|Q_h|}{T_h}$ and $\Delta S_c = +\dfrac{|Q_c|}{T_c}$.

Thus,

$$\Delta S_{\text{total}} = \Delta S_{\text{S}} + \Delta S_{\text{Carnot}} + \Delta S_h + \Delta S_c = 0 + 0 - \frac{83.3\text{ J}}{750\text{ K}} + \frac{0}{300\text{ K}}$$

$$= \boxed{-0.111\text{ J/K}}$$

(j) A decrease in total entropy is impossible.

P22.69 (a) Let state *i* represent the gas before its compression and state *f* afterwards, $V_f = \dfrac{V_i}{8}$. For a diatomic ideal gas, $C_V = \dfrac{5}{2}R$, $C_p = \dfrac{7}{2}R$, and $\gamma = \dfrac{C_p}{C_V} = 1.40$. Next,

$$P_i V_i^{\gamma} = P_f V_f^{\gamma}$$

$$P_f = P_i\left(\frac{V_i}{V_f}\right)^{\gamma} = P_i 8^{1.40} = 18.4 P_i$$

$$P_i V_i = nRT_i$$

$$P_f V_f = \frac{18.4 P_i V_i}{8} = 2.30 P_i V_i = 2.30 nRT_i = nRT_f$$

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so $T_f = 2.30T_i$

$$\Delta E_{\text{int}} = nC_V\Delta T = n\frac{5}{2}R\left(T_f - T_i\right) = \frac{5}{2}nR\left(1.30T_i\right) = \frac{5}{2}\left(1.30P_iV_i\right)$$

$$= \frac{5}{2}(1.30)\left(1.013\times10^5\ \text{N/m}^2\right)\left(0.120\times10^{-3}\ \text{m}^3\right) = 39.4\ \text{J}$$

Since the process is adiabatic, $Q = 0$ and $\Delta E_{\text{int}} = Q + W$ gives

$$W = \boxed{39.4\ \text{J}}$$

(b) The moment of inertia of the wheel is

$$I = \frac{1}{2}MR^2 = \frac{1}{2}(5.10\ \text{kg})(0.085\ 0\ \text{m})^2 = 0.018\ 4\ \text{kg}\cdot\text{m}^2$$

We want the flywheel to do work 39.4 J, so the work on the flywheel should be –39.4 J:

$$K_{\text{rot}\,i} + W = K_{\text{rot}\,f}$$

$$\frac{1}{2}I\omega_i^2 - 39.4\ \text{J} = 0$$

$$\omega_i = \left[\frac{2(39.4\ \text{J})}{0.018\ 4\ \text{kg}\cdot\text{m}^2}\right]^{1/2} = \boxed{65.4\ \text{rad/s} = 625\ \text{rev/min}}$$

(c) Now we want $W = 0.05K_{\text{rot}\,i}$:

$$39.4\ \text{J} = 0.05\left[\frac{1}{2}\left(0.018\ 4\ \text{kg}\cdot\text{m}^2\right)\omega_i^2\right]$$

$$\omega_i = \left(\frac{2(789\ \text{J})}{0.018\ 4\ \text{kg}\cdot\text{m}^2}\right)^{1/2} = \boxed{293\ \text{rad/s} = 2.79\times10^3\ \text{rev/min}}$$

P22.70 Like a refrigerator, an air conditioner has as its purpose the removal of energy by heat from the cold reservoir.

Its ideal COP is $\text{COP}_{\text{Carnot}} = \dfrac{T_c}{T_h - T_c} = \dfrac{280\ \text{K}}{20.0\ \text{K}} = 14.0.$

(a) Its actual COP is

$$0.400(14.0) = 5.60 = \frac{|Q_c|}{|Q_h| - |Q_c|} = \frac{|Q_c/\Delta t|}{|Q_h/\Delta t| - |Q_c/\Delta t|}$$

$$5.60\left|\frac{Q_h}{\Delta t}\right| - 5.60\left|\frac{Q_c}{\Delta t}\right| = \left|\frac{Q_c}{\Delta t}\right|$$

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$$5.60(10.0\text{ kW}) = 6.60\left|\frac{Q_c}{\Delta t}\right| \quad \text{and} \quad \left|\frac{Q_c}{\Delta t}\right| = \boxed{8.48\text{ kW}}$$

(b) $|Q_h| = W_{\text{eng}} + |Q_c|$:

$$\frac{W_{\text{eng}}}{\Delta t} = \left|\frac{Q_h}{\Delta t}\right| - \left|\frac{Q_c}{\Delta t}\right| = 10.0\text{ kW} - 8.48\text{ kW} = \boxed{1.52\text{ kW}}$$

(c) The air conditioner operates in a cycle, so the entropy of the working fluid does not change. The hot reservoir increases in entropy by

$$\frac{|Q_h|}{T_h} = \frac{(10.0\times10^3\text{ J/s})(3\,600\text{ s})}{300\text{ K}} = 1.20\times10^5\text{ J/K}$$

The cold room decreases in entropy by

$$\Delta S = -\frac{|Q_c|}{T_c} = -\frac{(8.48\times10^3\text{ J/s})(3\,600\text{ s})}{280\text{ K}}$$
$$= -1.09\times10^5\text{ J/K}$$

The net entropy change is positive, as it must be:

$$+1.20\times10^5\text{ J/K} - 1.09\times10^5\text{ J/K} = \boxed{1.09\times10^4\text{ J/K}}$$

(d) The new ideal COP is $\text{COP}_{\text{Carnot}} = \dfrac{T_c}{T_h - T_c} = \dfrac{280\text{ K}}{25\text{ K}} = 11.2.$

We suppose the actual COP is 0.400(11.2) = 4.48.

As a fraction of the original 5.60, this is $\dfrac{4.48}{5.60} = 0.800$, so the fractional change is to $\boxed{\text{drop by 20.0\%}}$.

P22.71 $e_C = 1 - \dfrac{T_c}{T_h} = \dfrac{W_{\text{eng}}}{|Q_h|} = \dfrac{W_{\text{eng}}/\Delta t}{|Q_h|/\Delta t}$: $\dfrac{|Q_h|}{\Delta t} = \dfrac{P}{(1 - T_c/T_h)} = \dfrac{PT_h}{T_h - T_c}$

$|Q_h| = W_{\text{eng}} + |Q_c|$: $\dfrac{|Q_c|}{\Delta t} = \dfrac{|Q_h|}{\Delta t} - \dfrac{W_{\text{eng}}}{\Delta t}$

$$\frac{|Q_c|}{\Delta t} = \frac{PT_h}{T_h - T_c} - P = \frac{PT_c}{T_h - T_c}$$

$|Q_c| = mc\Delta T$: $\dfrac{|Q_c|}{\Delta t} = \left(\dfrac{\Delta m}{\Delta t}\right)c\Delta T = \dfrac{PT_c}{T_h - T_c}$

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$$\frac{\Delta m}{\Delta t} = \frac{PT_c}{(T_h - T_c)c\Delta T}$$

$$\frac{\Delta m}{\Delta t} = \frac{(1.00\times10^9\ \text{W})(300\ \text{K})}{(200\ \text{K})(4\,186\ \text{J/kg}\cdot°\text{C})(6.00°\text{C})} = \boxed{5.97\times10^4\ \text{kg/s}}$$

P22.72 $e_C = 1 - \frac{T_c}{T_h} = \frac{W_{\text{eng}}}{|Q_h|} = \frac{W_{\text{eng}}/\Delta t}{|Q_h|/\Delta t}$: $\frac{|Q_h|}{\Delta t} = \frac{P}{1-(T_c/T_h)} = \frac{PT_h}{T_h - T_c}$

$$\frac{|Q_c|}{\Delta t} = \left(\frac{|Q_h|}{\Delta t}\right) - P = \frac{PT_c}{T_h - T_c}$$

But $|Q_c| = mc\Delta T$, where c is the specific heat of water.

Therefore, $\frac{|Q_c|}{\Delta t} = \left(\frac{\Delta m}{\Delta t}\right)c\Delta T = \frac{PT_c}{T_h - T_c}$

and $\frac{\Delta m}{\Delta t} = \boxed{\frac{PT_c}{(T_h - T_c)c\Delta T}}$.

P22.73 (a) For the isothermal process AB, the work on the gas is

$$W_{AB} = -P_A V_A \ln\left(\frac{V_B}{V_A}\right)$$

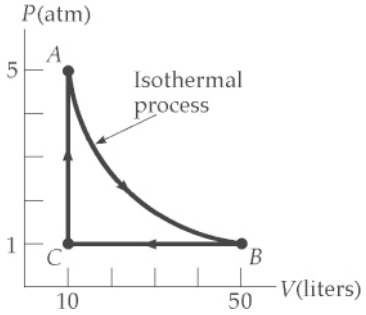
$$W_{AB} = -5(1.013\times10^5\ \text{Pa})(10.0\times10^{-3}\ \text{m}^3)\ln\left(\frac{50.0\ \text{L}}{10.0\ \text{L}}\right)$$

$$W_{AB} = -8.15\times10^3\ \text{J}$$

where we have used 1.00 atm = 1.013×10^5 Pa and 1.00 L = 1.00×10^{-3} m^3.

$$W_{BC} = -P_B\Delta V = -(1.013\times10^5\ \text{Pa})[(10.0 - 50.0)\times10^{-3}]\text{m}^3$$
$$= +4.05\times10^3\ \text{J}$$

$$W_{CA} = 0 \quad \text{and} \quad W_{\text{eng}} = -W_{AB} - W_{BC} = 4.10\times10^3\ \text{J} = \boxed{4.10\ \text{kJ}}$$



ANS. FIG. P22.73

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(b) Since AB is an isothermal process, $\Delta E_{\text{int},\,AB} = 0$

and $Q_{AB} = -W_{AB} = 8.15 \times 10^3$ J.

For an ideal monatomic gas, $C_V = \dfrac{3R}{2}$ and $C_P = \dfrac{5R}{2}$.

$$T_B = T_A = \frac{P_B V_B}{nR} = \frac{\left(1.013\times10^5\ \text{Pa}\right)\left(50.0\times10^{-3}\ \text{m}^3\right)}{R} = \frac{5.06\times10^3}{R}$$

Also, $T_C = \dfrac{P_C V_C}{nR} = \dfrac{\left(1.013\times10^5\ \text{Pa}\right)\left(10.0\times10^{-3}\ \text{m}^3\right)}{R} = \dfrac{1.01\times10^3}{R}$

$$Q_{CA} = nC_V\Delta T = 1.00\left(\frac{3}{2}R\right)\left(\frac{5.06\times10^3 - 1.01\times10^3}{R}\right) = 6.08\ \text{kJ}$$

so the total energy absorbed by heat is

$$Q_{AB} + Q_{CA} = 8.15\ \text{kJ} + 6.08\ \text{kJ} = \boxed{1.42\times10^4\ \text{J}}$$

(c) $Q_{BC} = nC_P\Delta T = \dfrac{5}{2}(nR\Delta T) = \dfrac{5}{2}P_B\Delta V_{BC}$

$$|Q_{BC}| = \left|\frac{5}{2}\left(1.013\times10^5\right)\left[(10.0-50.0)\times10^{-3}\right]\right| = \left|-1.01\times10^4\ \text{J}\right|$$

$$= \boxed{1.01\times10^4\ \text{J}}$$

(d) $e = \dfrac{W_{\text{eng}}}{|Q_h|} = \dfrac{W_{\text{eng}}}{Q_{AB}+Q_{CA}} = \dfrac{4.10\times10^3\ \text{J}}{1.42\times10^4\ \text{J}} = 0.289$ or $\boxed{28.9\%}$

(e) A Carnot engine operating between $T_{\text{hot}} = T_A = 5\,060/R$ and $T_{\text{cold}} = T_C = 1\,010/R$ has $e_C = 1 - T_c/T_h = 1 - 1/5 = \boxed{80.0\%}$. The efficiency of the cycle is much lower than that of a Carnot engine operating between the same temperature extremes.

P22.74 (a) The ideal gas at constant temperature keeps constant internal energy. As it puts out energy by work in expanding, it must take in an equal amount of energy by heat. Thus its entropy increases. Let P_i, V_i, and T_i represent the state of the gas before the isothermal expansion. Let P_C, V_C, and T_i represent the state after this process, so that $P_iV_i = P_CV_C$. Let P_i, $3V_i$, and T_f represent the state after the adiabatic compression.

Then $P_CV_C^{\gamma} = P_i\left(3V_i\right)^{\gamma}$

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Substituting $P_C = \dfrac{P_i V_i}{V_C}$

gives $P_i V_i V_C^{\gamma-1} = P_i\left(3^\gamma V_i^\gamma\right)$

Then $V_C^{\gamma-1} = 3^\gamma V_i^{\gamma-1}$ and $\dfrac{V_C}{V_i} = 3^{\gamma/(\gamma-1)}$

The work output in the isothermal expansion is

$$W = \int_i^C P\,dV = nRT_i \int_i^C V^{-1}\,dV$$

$$= nRT_i \ln\left(\frac{V_C}{V_i}\right) = nRT_i \ln\left(3^{\gamma/(\gamma-1)}\right) = nRT_i\left(\frac{\gamma}{\gamma-1}\right)\ln 3$$

This is also the input heat, so the entropy change is

$$\Delta S = \frac{Q}{T} = nR\left(\frac{\gamma}{\gamma-1}\right)\ln 3$$

Since $C_P = \gamma C_V = C_V + R$,

we have $(\gamma-1)C_V = R$, $C_V = \dfrac{R}{\gamma-1}$

and $C_P = \dfrac{\gamma R}{\gamma-1}$.

Then the result is $\boxed{\Delta S = nC_P \ln 3}$.

(b) The pair of processes considered here carries the gas from the initial state in P22.77 to the final state here. Entropy is a function of state. Entropy change does not depend on path. Therefore the entropy change in P22.77 equals $\Delta S_{\text{isothermal}} + \Delta S_{\text{adiabatic}}$ in this problem. Since $\Delta S_{\text{adiabatic}} = 0$, the answers to P22.77 and P22.74(a) must be the same.

P22.75 We recognize that $T_c = T_1$ and $T_h = T_2$, and $Q_c = 350$ J and $Q_h = -1\,000$ J.

$$\Delta S_{\text{hot}} = \frac{Q_h}{T_h} = -\frac{|Q_h|}{T_2} = \frac{-1000\text{ J}}{600\text{ K}}$$

$$\Delta S_{\text{cold}} = \frac{Q_c}{T_c} = \frac{|Q_c|}{T_1} = \frac{+750\text{ J}}{350\text{ K}}$$

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(a) $\Delta S_U = \Delta S_{\text{hot}} + \Delta S_{\text{cold}} = -\dfrac{1\,000\text{ J}}{600\text{ K}} - \dfrac{750\text{ J}}{350\text{ K}} = \boxed{0.476\text{ J/K}}$

(b) $e_C = 1 - \dfrac{T_1}{T_2} = 1 - \dfrac{350\text{ K}}{600\text{ K}} = 0.417$

$W_{\text{eng}} = e_C |Q_h| = 0.417(1\,000\text{ J}) = \boxed{417\text{ J}}$

(c) $\Delta W = W_C - W_{\text{real}}$

$= e_C|Q_h| - \left(|Q_h| - |Q_c|\right)$

$= \left(1 - \dfrac{T_c}{T_h}\right)|Q_h| - \left(|Q_h| - |Q_c|\right)$

$= \left(|Q_h| - \dfrac{T_c}{T_h}|Q_h|\right) - \left(|Q_h| - |Q_c|\right)$

$= |Q_c| - \dfrac{T_c}{T_h}|Q_h| = T_c\left(\dfrac{|Q_c|}{T_c} - \dfrac{|Q_h|}{T_h}\right)$

$= T_c \Delta S_U = T_1 \Delta S_U$

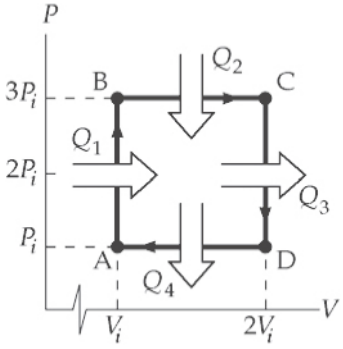
P22.76 At point A, $P_i V_i = nRT_i$ and $n = 1.00$ mol.

At point B, $3P_i V_i = nRT_B$ so $T_B = 3T_i$.

At point C, $(3P_i)(2V_i) = nRT_C$ and $T_C = 6T_i$.

At point D, $P_i(2V_i) = nRT_D$ so $T_D = 2T_i$.

The heat for each step in the cycle is found using $C_V = \dfrac{3R}{2}$ and $C_P = \dfrac{5R}{2}$:



ANS. FIG. P22.76

$Q_{AB} = nC_V(3T_i - T_i) = 3nRT_i$

$Q_{BC} = nC_P(6T_i - 3T_i) = 7.50nRT_i$

$Q_{CD} = nC_V(2T_i - 6T_i) = -6nRT_i$

$Q_{DA} = nC_P(T_i - 2T_i) = -2.50nRT_i$

(a) Therefore, $Q_{\text{entering}} = |Q_h| = Q_{AB} + Q_{BC} = 3nRT_i + 7.5nRT_i = \boxed{10.5nRT_i}$.

(b) $Q_{\text{leaving}} = |Q_c| = |Q_{CD} + Q_{DA}| = |-6nRT_i - 2.50nRT_i| = \boxed{8.50nRT_i}$

(c) Actual efficiency: $e = \dfrac{|Q_h| - |Q_c|}{|Q_h|} = \dfrac{10.5nRT_i - 8.5nRT_i}{10.5nRT_i} = \boxed{0.190}$

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(d) Carnot efficiency: $e_C = 1 - \frac{T_c}{T_h} = 1 - \frac{T_i}{6T_i} = \boxed{0.833}$

$\boxed{\text{The Carnot efficiency is much higher.}}$

P22.77 $$\Delta S = \int_i^f \frac{dQ}{T} = \int_i^f \frac{nC_P dT}{T} = nC_P \int_i^f T^{-1} dT = nC_P \ln T\Big|_{T_i}^{T_f} = nC_P\left(\ln T_f - \ln T_i\right)$$

$$= nC_P \ln\left(\frac{T_f}{T_i}\right)$$

$$\Delta S = nC_P \ln\left(\frac{PV_f}{nR}\frac{nR}{PV_i}\right) = nC_P \ln\left[\frac{P(3V_i)}{nR}\frac{nR}{PV_i}\right] = \boxed{nC_P \ln 3}$$

P22.78 (a) water: $T_{\text{water}} = 35.0°\text{F} \rightarrow \frac{5}{9}(35.0 - 32.0)°\text{C}$

$$\rightarrow (1.67 + 273.15)\ \text{K} = 274.82\ \text{K}$$

body: $T_{\text{body}} = 98.6°\text{F} \rightarrow \frac{5}{9}(98.6 - 32.0)°\text{C}$

$$\rightarrow (37.0 + 273.15)\ \text{K} = 310.15\ \text{K}$$

$$\Delta S_{\text{cold water}} = \int \frac{dQ}{T} = m_{\text{w}} c \times \int_{T_{\text{water}}}^{T_{\text{body}}} \frac{dT}{T} = m_{\text{w}} c \times \ln\left(\frac{T_{\text{body}}}{T_{\text{water}}}\right)$$

$$\Delta S_{\text{body}} = -\frac{|Q|}{T_{\text{body}}} = -\frac{m_{\text{w}} c\left(T_{\text{body}} - T_{\text{water}}\right)}{T_{\text{body}}}$$

$$\Delta S_{\text{system}} = \Delta S_{\text{cold water}} + \Delta S_{\text{body}}$$

$$= (0.454\ \text{kg})(4\ 186\ \text{J/kg}\cdot\text{K}) \times \ln\left(\frac{310.15}{274.82}\right)$$

$$-(0.454\ \text{kg})(4\ 186\ \text{J/kg}\cdot\text{K})\frac{(310.15 - 274.82)}{310.15} = \boxed{13.4\ \text{J/K}}$$

(b) Conservation of energy, $Q_{\text{hot}} = -Q_{\text{cold}}$, gives

$$m_w \cancel{c}\left(T_F - T_{\text{water}}\right) = -m_{\text{Ath}} \cancel{c}\left(T_F - T_{\text{body}}\right)$$

$$m_w\left(T_F - T_{\text{water}}\right) = -m_{\text{Ath}}\left(T_F - T_{\text{body}}\right)$$

$$m_w T_F - m_w T_{\text{water}} = -m_{\text{Ath}} T_F + m_{\text{Ath}} T_{\text{body}}$$

$$\left(m_w + m_{\text{Ath}}\right) T_F = m_w T_{\text{water}} + m_{\text{Ath}} T_{\text{body}}$$

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Solving for T_F,

$$T_F = \frac{m_w T_{\text{water}} + m_{\text{Ath}} T_{\text{body}}}{m_w + m_{\text{Ath}}}$$

$$= \frac{(0.454\ \text{kg})(274.82\ \text{K}) + (70.0\ \text{kg})(310.15\ \text{K})}{0.454\ \text{kg} + 70.0\ \text{kg}}$$

$$= 309.92\ \text{K} = \boxed{310\ \text{K}}$$

(c) $\Delta S = \Delta S'_{\text{ice water}} + \Delta S'_{\text{body}}$

$$= m_{\text{w}} c \times \ln\left(\frac{T_F}{T_{\text{water}}}\right) + m_{\text{Ath}} c \times \ln\left(\frac{T_F}{T_{\text{body}}}\right)$$

$$= (0.454\ \text{kg})(4\ 186\ \text{J/kg}\cdot\text{K})\ln\left(\frac{309.92}{274.82}\right)$$

$$+ (70.0\ \text{kg})(4\ 186\ \text{J/kg}\cdot\text{K})\ln\left(\frac{309.92}{310.15}\right)$$

$$= \boxed{11.1\ \text{J/K}}$$

(d) Smaller by less than 1%

P22.79 (a) $W = \int_{V_i}^{V_f} P dV = nRT \int_{V_i}^{2V_i} \frac{dV}{V} = (1.00) RT \ln\left(\frac{2V_i}{V_i}\right) = \boxed{RT \ln 2}$

(b) Yes.

(c) No. The second law refers to an engine operating in a cycle, whereas this problem involves only a single process.

P22.80 When energy enters a substance by heat, we describe the process with Equation 20.4, $Q = mc\Delta T$. This is a reversible process; if energy leaves the substance, the temperature drops down again. Therefore, the entropy change for one of the samples of water is

$$\Delta S = \int \frac{dQ}{T} = \int_{T_i}^{T_f} \frac{mcdT}{T} = mc \ln\left(\frac{T_f}{T_i}\right)$$

Consequently, the entropy change for both samples of water is

$$\Delta S_{\text{total}} = \Delta S_{\text{hot}} + \Delta S_{\text{cold}}$$

$$= mc \ln\left(\frac{T_f}{T_{hi}}\right) + mc \ln\left(\frac{T_f}{T_{ci}}\right) = mc \ln\left[\left(\frac{T_f}{T_{hi}}\right)\left(\frac{T_f}{T_{ci}}\right)\right]$$

$$(1.00\ \text{kg})(4\ 186\ \text{J/kg}\cdot°\text{C})\ln\left[\left(\frac{293\ \text{K}}{303\ \text{K}}\right)\left(\frac{293\ \text{K}}{283\ \text{K}}\right)\right] = 4.88\ \text{J/K}$$

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This is *not* zero. While the statements about energy transfer by heat are true, the mixing process is irreversible. After the water has come to equilibrium, it will not spontaneously separate again into warm and cool water. Therefore, there is an entropy increase of the mixture during this irreversible process.

Challenge Problems

P22.81 (a) Given: $P_A = \boxed{25.0 \text{ atm}}$ and $P_C = \boxed{1.00 \text{ atm}}$

Use the equation of state for an ideal gas:

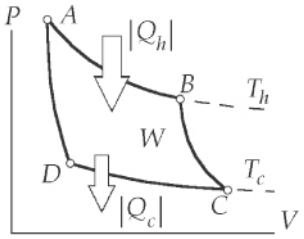
$$V = \frac{nRT}{P}$$

$$V_A = \frac{1.00\left(8.314 \text{ J/mol}\cdot\text{K}\right)(600 \text{ K})}{25.0\left(1.013\times10^5 \text{ Pa}\right)} = \boxed{1.97\times10^{-3} \text{ m}^3}$$

$$V_C = \frac{1.00\left(8.314 \text{ J/mol}\cdot\text{K}\right)(400 \text{ K})}{1.013\times10^5 \text{ Pa}} = \boxed{32.8\times10^{-3} \text{ m}^3}$$

Since AB is isothermal, $P_A V_A = P_B V_B$,

and since BC is adiabatic, $P_B V_B^{\gamma} = P_C V_C^{\gamma}$.



ANS. FIG. P22.81

Combining these expressions,

$$V_B = \left[\left(\frac{P_C}{P_A}\right)\frac{V_C^{\gamma}}{V_A}\right]^{1/(\gamma-1)}$$

$$= \left[\left(\frac{1.00 \text{ atm}}{25.0 \text{ atm}}\right)\left(\frac{\left(32.8\times10^{-3} \text{ m}^3\right)^{1.40}}{1.97\times10^{-3} \text{ m}^3}\right)\right]^{(1/0.400)}$$

$$= \boxed{11.9\times10^{-3} \text{ m}^3}$$

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Similarly,

$$V_D = \left[\left(\frac{P_A}{P_C}\right)\frac{V_A^{\gamma}}{V_C}\right]^{1/(\gamma-1)}$$

$$= \left[\left(\frac{25.0 \text{ atm}}{1.00 \text{ atm}}\right)\left(\frac{\left(1.97\times10^{-3} \text{ m}^3\right)^{1.40}}{32.8\times10^{-3} \text{ m}^3}\right)\right]^{(1/0.400)}$$

$$= \boxed{5.44\times10^{-3} \text{ m}^3}$$

Since AB is isothermal, $P_A V_A = P_B V_B$

and $P_B = P_A\left(\frac{V_A}{V_B}\right) = (25.0 \text{ atm})\left(\frac{1.97\times10^{-3} \text{ m}^3}{11.9\times10^{-3} \text{ m}^3}\right) = \boxed{4.14 \text{ atm}}$

Also, CD is an isothermal and

$$P_D = P_C\left(\frac{V_C}{V_D}\right) = (1.00 \text{ atm})\left(\frac{32.8\times10^{-3} \text{ m}^3}{5.44\times10^{-3} \text{ m}^3}\right) = \boxed{6.03 \text{ atm}}$$

(b) Energy is added by heat to the gas during the process AB. For the isothermal process, $\Delta E_{\text{int}} = 0,$ and the first law gives

$$Q_{AB} = -W_{AB} = nRT_h \ln\left(\frac{V_B}{V_A}\right)$$

or

$$|Q_h| = Q_{AB} = (1.00 \text{ mol})\left(8.314 \text{ J/mol}\cdot\text{K}\right)(600 \text{ K})\ln\left(\frac{11.9 \text{ atm}}{1.97 \text{ atm}}\right)$$

$$= 8.97 \text{ kJ}$$

Then, from $e = \frac{W_{\text{eng}}}{|Q_h|}$, the net work done per cycle is

$$W_{\text{eng}} = e_c |Q_h| = 0.333(8.97 \text{ kJ}) = \boxed{2.99 \text{ kJ}}$$

P22.82 The quantity of gas is

$$n = \frac{P_A V_A}{RT_A} = \frac{\left(100\times10^3 \text{ Pa}\right)\left(500\times10^{-6} \text{ m}^3\right)}{\left(8.314 \text{ J/mol}\cdot\text{K}\right)(293 \text{ K})} = 0.020\ 5 \text{ mol}$$

(a) In process $A\rightarrow B$,

$$P_B = P_A\left(\frac{V_A}{V_B}\right)^{\gamma} = \left(100\times10^3 \text{ Pa}\right)(8.00)^{1.40} = 1.84\times10^6 \text{ Pa}$$

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$$T_B = \frac{P_B V_B}{nR} = \frac{(1.84\times10^6\ \text{Pa})(500\times10^{-6}\ \text{m}^3/8.00)}{(0.020\ 5\ \text{mol})(8.314\ \text{J/mol}\cdot\text{K})} = 673\ \text{K}$$

$$V_A/V_B = 8.00 \quad \rightarrow \quad V_B = V_A/8.00 = 500/8 \ = \ 62.5\ \text{cm}^3$$

State C:

$$V_C = V_B$$

$$P_C = \frac{nRT_C}{V_C} = \frac{(0.020\ 5\ \text{mol})(8.314\ \text{J/mol}\cdot\text{K})(1\ 023\ \text{K})}{62.5\times10^{-6}\ \text{m}^3}$$

$$= 2.79\times10^6\ \text{Pa}$$

State D:

$$V_D = V_A$$

In process $C\rightarrow D$:

$$P_D = P_C\left(\frac{V_C}{V_D}\right)^{\gamma} = (2.79\times10^6\ \text{Pa})\left(\frac{1}{8.00}\right)^{1.40} = 1.52\times10^5\ \text{Pa}$$

$$T_D = \frac{P_D V_D}{nR} = \frac{(1.52\times10^5\ \text{Pa})(500\times10^{-6}\ \text{m}^3)}{(0.020\ 5\ \text{mol})(8.314\ \text{J/mol}\cdot\text{K})} = 445\ \text{K}$$

TABLE P22.82(a) tabulates these results:

	T (K)	P (kPa)	V (cm^3)
A	293	100	500
B	673	1.84×10^3	62.5
C	1023	2.79×10^3	62.5
D	445	152	500

TABLE P22.82(a)

(b) In the adiabatic process $A\rightarrow B$, $Q = 0$,

$$\Delta E_{\text{int},\,A\rightarrow B} = \frac{5}{2}nR(T_B - T_A)$$

$$= \frac{5}{2}(0.020\ 5\ \text{mol})(8.314\ \text{J/mol}\cdot\text{K})(673\ \text{K} - 293\ \text{K})$$

$$= 162\ \text{J}$$

and $\Delta E_{\text{int},\,AB} = 162\ \text{J} = Q - W_{\text{out}} = 0 - W_{\text{out}} \quad \rightarrow \quad W_{AB} = -162\ \text{J}$

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In the isovolumetric process $B \rightarrow C$, $W = 0$

$$\Delta E_{\text{int, } B\rightarrow C} = \frac{5}{2} nR\left(T_C - T_B\right)$$

$$= \frac{5}{2}(0.020\ 5\ \text{mol})\left(8.314\ \text{J/mol}\cdot\text{K}\right)(1\ 023\ \text{K} - 673\ \text{K})$$

$$= 149\ \text{J}$$

$$\Delta E_{\text{int, } B\rightarrow C} = 149\ \text{J} = Q - W_{\text{out}} = Q - 0 \quad \rightarrow \quad Q_{BC} = 149\ \text{J}$$

In the adiabatic process $C \rightarrow D$, $Q = 0$

$$\Delta E_{\text{int, } C\rightarrow D} = \frac{5}{2} nR\left(T_D - T_C\right)$$

$$= \frac{5}{2}(0.020\ 5\ \text{mol})\left(8.314\ \text{J/mol}\cdot\text{K}\right)(445\ \text{K} - 1\ 023\ \text{K})$$

$$= -246\ \text{J}$$

$$\Delta E_{\text{int, } C\rightarrow D} = -246\ \text{J} = Q - W_{\text{out}} = 0 - W_{\text{out}} \quad \rightarrow \quad W_{CD} = 246\ \text{J}$$

For the entire cycle, $\Delta E_{\text{int, net}} = \frac{5}{2} nR\Delta T = 0$:

$$W_{\text{eng}} = -162\ \text{J} + 0 + 246.3\ \text{J} + 0 = 84.3\ \text{J}$$

$$\Delta E_{\text{int}} = Q_{\text{net}} + W_{\text{eng}} = 0 \quad \rightarrow \quad Q_{\text{net}} = -W_{\text{eng}} = 84.3\ \text{J}$$

TABLE P22.82(b) tabulates these results:

	Q	W_{eng}	ΔE_{int}
$A \rightarrow B$	0	–162	162
$B \rightarrow C$	149	0	149
$C \rightarrow D$	0	246	–246
$D \rightarrow A$	–65.0	0	–65.0
$ABCD$	84.3	84.3	0

TABLE P22.82(b)

(c) From B→C, the input energy is $Q_h = \boxed{149\ \text{J}}$.

(d) From D→A, the energy exhaust is $|Q_c| = \boxed{65.0\ \text{J}}$.

(e) From ABCDA, $W_{\text{eng}} = \boxed{84.3\ \text{J}}$.

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(f) The efficiency is: $e = \frac{W_{\text{eng}}}{Q_h} = \frac{84.3\ \text{J}}{149\ \text{J}} = \boxed{0.565}$

(g) Let f represent the angular speed of the crankshaft. Then $\frac{f}{2}$ is the frequency at which we obtain work in the amount of 84.3 J/cycle:

$$1\ 000\ \text{J/s} = \left(\frac{f}{2}\right)(84.3\ \text{J/cycle})$$

$$f = \frac{2\ 000\ \text{J/s}}{84.3\ \text{J/cycle}} = 23.7\ \text{rev/s} = \boxed{1.42 \times 10^3\ \text{rev/min}}$$

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ANSWERS TO EVEN-NUMBERED PROBLEMS

P22.2 (a) 0.25 or 25%; (b) $|Q_C|/|Q_h| = 3/4$

P22.4 13.7°C

P22.6 (a) 29.4 L/h; (b) 185 hp; (c) 527 N · m; (d) 1.91×10^5 W

P22.8 (a) 24.0 J; (b) 144 J

P22.10 (a) 7.69×10^8 J; (b) 5.67×10^8 J

P22.12 (a) 2.65×10^7 J; (b) 3.20

P22.14 0.540 or 54.0%

P22.16 The efficiency of a Carnot engine operating between these temperatures is 6.83%. Therefore, there is no way that the inventor's engine can have an efficiency of 0.110 = 11.0%.

P22.18 (a) $P\Delta t\left(\frac{T_h}{T_h - T_c}\right)$; (b) $P\Delta t\left(\frac{T_c}{T_h - T_c}\right)$

P22.20 (a) See P22.20(a) for the full solution; (b) See P22.20(b) for the full solution.

P22.22 72.2 J

P22.24 0.330 or 33.0%

P22.26 (a) 0.300; (b) 1.40×10^{-3} K^{-1}; (c) -2.00×10^{-3} K^{-1}; (d) No. The derivative in part (c) depends only on T_h

P22.28 (a) 5.12%; (b) 5.27 TJ/h; (c) As fossil-fuel prices rise, this way to use solar energy will become a good buy.

P22.30 (a) See P22.30(a) for full explanation; (b) $1 - \frac{T_c}{T_h}$; (c) The combination of reversible engines is itself a reversible engine so it has the Carnot efficiency. No improvement in net efficiency has resulted; (d) $T_i = \frac{1}{2}(T_h + T_c)$;
(e) $T_i = (T_h T_c)^{1/2}$

P22.32 (a) See TABLE P22.32(a); (b) See TABLE P22.32(b); (c) 23.7%; (d) 23.7%

P22.34 23.1 mW

P22.36 (a) 51.2%; (b) 36.2%

P22.38 See P22.38 for the full derivation

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P22.40 (a) See TABLE P22.40; (b) 2 heads and 2 tails

P22.42 −610 J/K

P22.44 717 J/K

P22.46 2.70 kJ/K

P22.48 (a) 5.76 J/K; (b) no change in temperature

P22.50 244 J/K

P22.52 1 W/K

P22.54 (a) $\Delta S_h = -\frac{Q}{T_h}$; (b) $\Delta S_c = -\frac{Q}{T_c}$; (c) $Q\left(\frac{1}{T_c} - \frac{1}{T_h}\right)$

P22.56 +1.18 J/K

P22.58 (a) 0.268; (b) 0.423

P22.60 8.36×10^6 J/K · s

P22.62 (a) 2.93; (b) $(\text{COP})_{\text{refrigerator}}$; (c) with EER 5, \$510, with EER 10, \$255; Thus, the cost for air conditioning is half as much for an air conditioner with EER 10 compared with an air conditioner with EER 5.

P22.64 (a) 2.49 kJ; (b) 1.50 kJ; (c) –990 J

P22.66 (a) $\frac{T_c}{T_h - T_c}$; (b) smaller; (c) 14.6

P22.68 (a) 214 J and 64.3 J; (b) 35.7 J and 35.7 J. The total energy the firebox puts out equals to the total energy transferred to the environment; (c) The net flow of energy by heat from the cold to the hot reservoir without work input is possible; (d) $W_{\text{eng S}} = 233$ J, $\left|Q_{h,\text{S}}\right| = 333$ J; (e) 83.3 J; (f) 83.3 J; (g) 0; (h) The output of 83.8 J of energy from the heat engine by work in a cyclic process without any exhaust by heat is impossible; (i) −0.111 J/K; (j) A decrease in total entropy is impossible.

P22.70 (a) 8.48 kW; (b) 1.52 kW; (c) 1.09×10^4 J/K; (d) drop by 20.0%

P22.72 $\frac{PT_c}{(T_h - T_c)c\Delta T}$

P22.74 (a) $\Delta S = nC_p \ln 3$; (b) The pair of processes considered here carries the gas from the initial state in P22.77 to the final state here. Entropy is a function of state. Entropy change does not depend on path. Therefore, the entropy change in P22.77 equals $\Delta S_{\text{isothermal}} + \Delta S_{\text{adiabatic}}$ in this problem. Since $\Delta S_{\text{adiabatic}} = 0$, the answers to P22.77 and P22.74(a) must be the same.

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P22.76 (a) 10.5 nRT_i; (b) 8.50 nRT_i; (c) 0.190; (d) 0.833; The Carnot efficiency is much higher.

P22.78 (a) 13.4 J/K; (b) 310 K; (c) 11.1 J/K; (d) smaller by less than 1%

P22.80 The computed change in entropy is 4.88 J/K, which is *not* zero. While the statements about energy transfer by heat are true, the mixing process is irreversible. After the water has come to equilibrium, it will not spontaneously separate again into warm and cool water. Therefore, there is an entropy increase of the mixture during the irreversible process.

P22.82 (a) See TABLE P22.82(a); (b) See TABLE P22.82(b); (c) 149 J; (d) 65.0 J; (e) 84.3 J; (f) 0.565; (g) 1.42×10^3 rev/min

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